

 **DTU Compute**
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Signal to noise properties in 4D hyperspectral X-ray data sets

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Preface

This Bachelor's thesis was completed at the Department of Applied Mathematics and Computer Science and in collaboration with the Department of Physics at the Technical University of Denmark to satisfy the criteria for obtaining a Bachelor of Science degree in Mathematics and Technology. The credit for this project is 20 ECTS.

Kongens Lyngby, June 9, 2023

A handwritten signature in black ink, reading "Ida Puggaard". The signature is written in a cursive style with a large, stylized 'I' and 'P'.

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Abstract

This project investigates the impact of different acquisition parameters, such as number of projections and acquisitions repetitions, on reconstruction quality when the photon count is kept constant in hyper spectral X-ray imaging.

Two distinct figure of merits (FOM) is identified and implemented to assess the quality of reconstructions obtained from a 2D hyper spectral data set. One FOM is based on Root Mean Square Error (RMSE) and the other is based on the utilisation of attenuation curves while using different methods for pixel sampling and error calculations. The two FOM's yielded different candidates for optimal acquisition parameters, due to the merits capturing different characteristics from the reconstruction. The RMSE based FOM found 420 projections and 6 acquisition repetitions to be optimal scan conditions while the energy curve based FOM found 36, 45, 70, and 90 projections with 70, 56, 36, and 28 repetitions respectively to be optimal.

The findings provided valuable insights into determining optimal acquisition configurations and demonstrate the significance of both implemented FOMs in assessing reconstruction quality. The main work of the project was done using the Core Imaging Library (CIL) python library.

Contents

Preface	i
Acknowledgements	iii
Abstract	v
Contents	vii
1 Introduction	1
2 Fundamentals of X-ray Imaging	3
2.1 X-rays	3
2.2 X-ray Interaction with Matter	4
2.3 Detection of X-rays	5
2.4 Radon Transformation	6
2.5 Computed Tomography	9
2.6 Signal Quality	11
3 Reconstruction Methods	13
3.1 Rays through Pixels	13
3.2 Setting up the System Matrix	14
3.3 Iterative Reconstruction Methods	15
3.4 Regularisation	16
4 Data Processing Techniques	21
4.1 Data Correction	21
4.2 Removing Dead/Hot Pixels	21
4.3 Detector Gap Interpolation	22
4.4 Ring Remover	23
5 Figure of Merit	25
5.1 Figure of Merit Based on a Reconstructed Image	25
5.2 Figure of Merit Based on Energy Curves	25
6 Preliminary Study	29
6.1 Data	29
6.2 Preprocessing of Data	33
6.3 Ground Truth Reconstruction	34
6.4 Choosing of an Optimal Regularisation Parameter	36
6.5 Best Acquisition Parameters	40

7	Regularisation Along the Energy Dimension	43
7.1	Construction of Energy Curve	43
7.2	Setting up Regularisation in the Energy Dimension	44
7.3	Different Choices of β	44
8	Data set with more projections	47
8.1	Design of the Data Set	47
8.2	Ground Truth Reconstruction	51
9	Experimental Study and Results	57
9.1	Outline of Desired Study	57
9.2	Regularisation Parameter Sweep	57
9.3	Acquisition Experiment Using RMSE as FOM	58
9.4	Acquisition Experiment Using the Energy Curves as FOM	60
10	Discussion	65
10.1	Discussion of Results	65
10.2	Further work	67
11	Conclusion	69
	Bibliography	71

CHAPTER 1

Introduction

Hyperspectral X-ray image acquisition provides a new dimension of information in computed tomography (CT) measurements and has become an increasingly popular technique for analysing the properties of materials. In this project will hyperspectral be used to describe imaging utilising approximately 100 energy channels which makes it a useful tool for accurate material identification in fields such as security screening. Unlike conventional X-ray imaging, new photon-counting detectors measure energy-resolved data from which energy-resolved images can be reconstructed. As the energy resolution is increased, more information is recorded to distinguish between materials. However, each energy-channel receives few photons leading to low image quality in each reconstructed energy-channel. This trade-off remains under explored, and the presented project will investigate this and seek to determine a better understanding of the optimal scan conditions for exploiting the energy dependence of the attenuation of materials for better classification.

This project evolves around investigating the impact of different acquisition parameters such as the number of projections and acquisition time on the noise properties in 4D hyperspectral X-ray projections for tomographic reconstruction. The primary data worked with in this project will emulate airport baggage screenings, where correct classification of materials is essential to ensure safety. In airport baggage scanners such as the one seen in figure 1.1 is it important to keep scanning time to a minimum, since many bags will have to be investigated in a short period of time. It is therefore valuable to investigate optimal acquisition configuration that yields acceptable reconstructions, while maintaining the photon budget.



Figure 1.1: X-ray scanner used in airport security.

The goal is to identify and implement distinct figures of merit (FOM) that can effectively assess the quality of reconstructions obtained under varying acquisition conditions while maintaining a constant photon count. By designing and employing such figures of merit, valuable insights can be gathered to determine optimal acquisition configurations that yield superior reconstruction outcomes. The proposed figures of merit will be evaluated based on their discriminating capabilities in identifying superior reconstructions, as well as their efficacy in facilitating the determination of optimal acquisition parameters.

The used data will mainly be processed and reconstruction with The Core Imaging Library (CIL) python framework (<https://ccpi.ac.uk/cil/>). The project will therefore also focus on developing a workflow for the processing of real spectral CT data within the framework, for future uses to gather inspiration from. The code developed throughout the course of the project will be published on GitHub: https://github.com/IdaPug/Bachelor_project_2023

CHAPTER 2

Fundamentals of X-ray Imaging

This chapter will describe the fundamentals of X-ray imaging and aims to give a simple outline on how data arises from scanning of objects using X-rays. The chapter is based on the following literature: [1], [2], [3], [4], [5], [6], [7], [8], [9] [10]

2.1 X-rays

X-rays are, similarly with the visible light, a type of electromagnetic wave, where X-rays are high frequency waves compared to the visible light. X-rays are known for their ability to penetrate most objects, and is therefore widely used for analysing properties of materials. The electronic waves' ability to penetrate different materials are dependent on the waves' energy. X-ray waves can range in wavelength between 0.1 Å- 100 Å. Shorter waves will have a higher energy and therefore be able to transmit through thicker objects than waves with a longer wavelength. The relationship between the X-ray wavelength λ and energy, E can be described using:

$$E = hv = \frac{hc}{\lambda}, \quad (2.1)$$

where, h is Planck's constant (6.63×10^{-34} Js), v is the frequency, and c is the speed of light (3×10^8 m/s).

2.1.1 Generation of X-rays

X-rays are generated in a high energy process when fast-moving electrons react with a solid target and thereby undergo a sudden deceleration called "bremsstrahlung" or "braking radiation". As the electrons slow down, they release some of their energy as X-ray photons. When an electron strikes the solid target two types of interactions can happen. In the case where a high-energy electron collides directly with a nucleus, the result is the loss of all its kinetic energy, and the emission of a photon with equal energy. The other case is when the electron passes the nucleus of an anode atom, and the result is the loss of some energy and the emission of a photon carrying that energy. The energy loss and photon energy decrease as the distance between the electron and nucleus increases.

In a laboratory X-rays are generated in a vacuum tube where electrons are released and accelerated toward an anode via a high-voltage electrostatic field. The anode is typically made of tungsten, copper, or silver and the choice depends on the energy

spectrum the X-ray source emits. The interactions between high-energy electrons and shell electrons of the anode material are significant, as they can ionise the atoms of the anode material by ejecting a k-shell electron. This process leads to the release of energy in the form of X-ray radiation when the k-shell vacancy is filled. The X-ray energy emitted from shell electrons has a narrow energy spectrum due to the fixed binding energies of the target material, which appears as spikes in the spectrum. These energy spikes are known as characteristic X-rays, which are distinctive for the anode material. The choice of anode material is therefore essential to how the energy spectrum will look, and the choice of material typically aims to place characteristics peaks central in the used range.

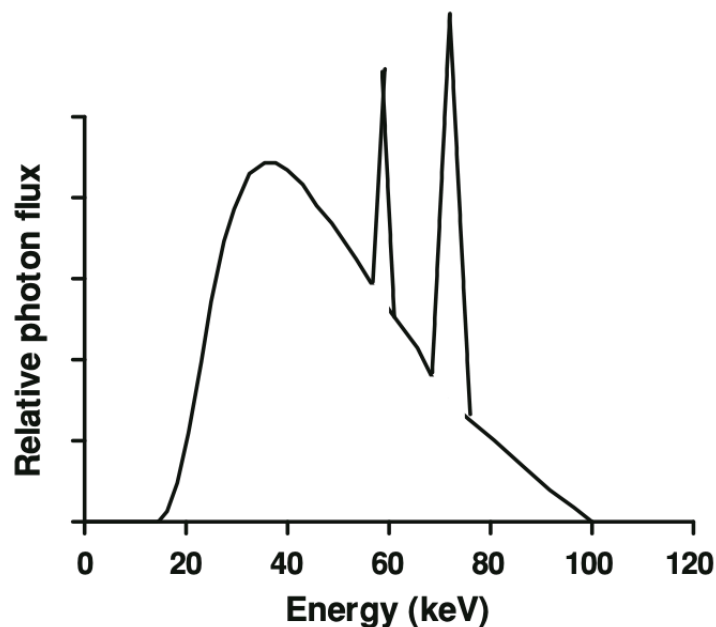


Figure 2.1: Typical X-ray spectrum of a tube with a tungsten anode operated at 100 kV. The graph have two peaks in the photon flux due to characteristic X-rays. Source [1].

2.2 X-ray Interaction with Matter

When X-rays pass through matter will it either interacts with an atom or passes through it without any interaction. The photon interacts via scattering or absorption and will deviate from its path or disappear. The result of this is that a lower number of photons will be detected on the other side; this is called attenuation. In 1729, Pierre Bouguer was the first to describe this phenomenon while studying light absorption. Johann Heinrich Lambert later noted in 1790 that the absorption of the material is directly proportional to its thickness. In 1852, August Beer discovered that the concentration of attenuating

species in a material sample is directly proportional to its light absorbance. All this led to Lambert-Beers law which can be derived as a solution to ordinary differential equations governing X-ray attenuation. Looking at the X-ray propagated along a line l . Lambert-Beer's law describing mono-chromatic attenuation, which is attenuation derived from radiation of a single wavelength, for non-homogeneous material along the line is then:

$$I(l) = I_0 e^{-\int_L \mu(x) dl} \quad (2.2)$$

I_0 is the incident flux, $I(l)$ is called the intensity and $\mu(l)$ is the linear attenuation coefficient though the material along the line l . $\mu(l)$ is a continuously variable which is integrated along the line though the material.

Rather than looking at just the intensity, it can be more useful to measure the change in X-ray photons passing through objects compared to having no material. We can therefore rewrite eq: (2.2) to:

$$\frac{I(l)}{I_0} = e^{-\int_L \mu(l) dl}, \quad (2.3)$$

where $\frac{I(l)}{I_0}$ is called the transmission and describes what ratio of the initial photons have passed through the material and will therefore be detected on the other side.

2.3 Detection of X-rays

Detection of X-rays can either be an indirect or direct process. In the an indirect approach the intensities of the rays are detected in a two-step progress. Initially, X-ray photons are absorbed by a scintillator and transformed into visible light. This component in the detector has the highest impact on image quality. The reason for this is that light tends to scatter as it passes through the scintillator, causing it to spread across neighboring pixels in the CCD. Since the amount of light diffusion is directly proportional to the distance it needs to traverse, a thin scintillator can provide better resolution than a thick one, however, it will absorb a lower amount of the incident radiation and thus produce a low signal-to-noise ratio (SNR). After the scintillator, the produced visible light is linked to a charge-coupled device (CCD). The CCD is a chip consisting of numerous small light-sensitive sensors, or pixels, arranged on a large grid. It operates as an analog device, where the photons that hit the sensors are transformed into electric charges, known as photoelectrons. The amount of charge collected by a sensor is proportional to the number of photons it receives.

2.3.1 Photon-counting Detectors and Multi Energy Detection

An X-ray beam is not mono-chromatic and the attenuation coefficient of different materials is thereby energy-dependent. Therefore in recent years, experimental X-ray systems have introduced photon-counting detectors (PCDs) capable of spectral imaging. PCDs is a type of direct detector and employ a direct conversion technique for X-ray detection, eliminating the need for a scintillator layer. Instead, the semiconductor detector material directly transforms X-ray photons into electron-hole pairs. By applying a bias voltage across the semiconductor, electrons migrate towards the anode where they are gathered, generating an electronic signal. Since PCDs count the number of individual photons that exceed a specified energy level, these detectors can register the energy of each detected photon. This leads to multi-energy X-ray imaging which requires the acquisition of two or more measurements with different energy spectra.

Multi-energy acquisition adds a new dimension to the captured data where each data-point has multiple absorption values corresponding to the energy levels emitted by the X-ray source. The intensity of a beam with energy E can be expressed via Lambert-Beers law 2.2 as:

$$I(l, E) = I_0(E)e^{-\int_L \mu(l, E)dl} \quad (2.4)$$

PCDs offer numerous advantages compared to traditional detectors. By exploiting the different attenuation of materials at various X-ray energies, it enables the classification of materials beyond what can be achieved with conventional detectors. However the separation of photons into energy channels causes each energy channel to receive fewer photons leading to lower image quality in each reconstructed energy channel, and some energy channels might even be useless due to the low amount of captured signal. The introduction of multi spectral X-ray techniques can also have positive effects such as lowering the risk of beam hardening, where X-rays hardened throughout the object of interest.

2.4 Radon Transformation

The Radon transformation is a forward model, named after the Austrian mathematician Johann Radon. Forward models are used to model how the captured data arises from the scanning of the object of interest. The Radon Transformation describes how the projection data arises in the ideal scenario. The transform takes a signal and produces a set of line integrals, which represent the projection of the image along different angles.

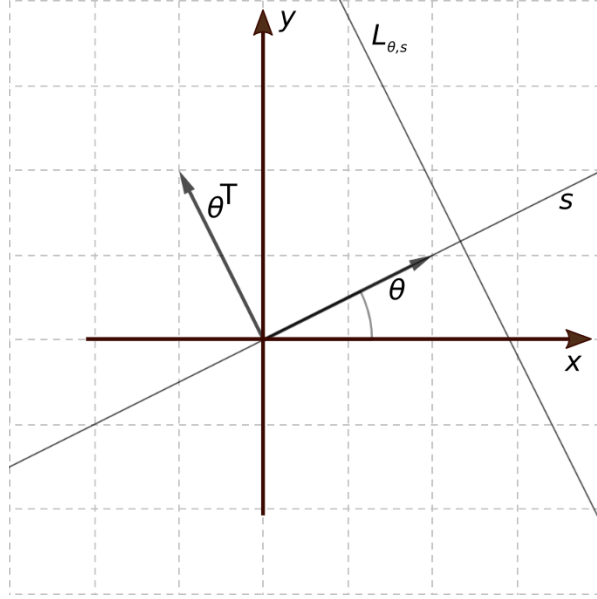


Figure 2.2: Geometry of the line represented by eq: (2.6). Figure made as a modification from source: [3], page 56.

2.4.1 Describing a Line

After defining the origin as the rotation axis of the object of interest, and given an angle θ , we can define the orthogonal unit direction vectors, θ and $\theta^\top$ as:

$$\theta = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \quad \text{and} \quad \theta^\top = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} \quad (2.5)$$

We then define the axis s which is rotated by the angle θ with respect to the x -axis with unit direction θ . We then denote the line $L_{\theta,s}$ as the line orthogonal to the unit direction θ and with signed distance s to the origin. An illustration of the line $L_{\theta,s}$ in a normal coordinate system can be seen in figure 3.1. Using the described notation a line in the plane can be formulated using its normal form:

$$L_{\theta,s} = \{(x, y) \mid x \cos \theta + y \sin \theta = s\} \quad (2.6)$$

All lines in the plane can be described using this notation. It is important to note that $L_{\theta+180^\circ,s} = L_{\theta,s}$, since s runs from $s \in [-p, p]$, but it is a convention to use $\theta \in [0^\circ, 360^\circ]$, so that will also be done in this project.

2.4.2 Definition of the Radon Transformation

The Radon transformation is defined as the projection that describes all the line integrals along all $L_{\theta,s}$ at a fixed θ as:

$$p_\theta(s) = \int_{L_{\theta,s}} f(x, y) d\ell, \quad (2.7)$$

where $s \in [-p, p]$ and p is the radius of a disk in the plane completely covering the objects of interest. The Radon transformation can then be formulated as

$$[\mathcal{R}f](\theta, s) = p_\theta(s) = \int_{L_{\theta,s}} f(x, y) d\ell, \quad (2.8)$$

for $\theta \in [0^\circ, 360^\circ]$ and $s \in [-p, p]$.

2.4.3 Relationship with The Lambert-Beer law

The Radon transformation is a mathematical construction and does not describe how the attenuated values arise from the line integrals, while the Lambert-Beer law formulated in eq: (2.2) describes how the transmission values arise as line integrals. Lambert-Beer's law along the same line parameterised in section 2.4.1 denoted $L_{\theta,s}$ is:

$$I_{\theta,s} = I_0 \exp \left(- \int_{L_{\theta,s}} \mu(x, y) d\ell \right) \quad (2.9)$$

This can then be linked to the radon transformation seen in equation 2.8 using the identifiers:

$$\begin{aligned} f(x, y) &= \mu(x, y) \\ p_\theta(s) &= -\log \left(\frac{I_{\theta,s}}{I_0} \right) \end{aligned} \quad (2.10)$$

2.4.4 The Sinogram

The output of the radon transformation $[\mathcal{R}f](\theta, s)$ can be visualised in what is called a sinogram by stacking all the projections $p_\theta(s)$ into a 2D data set $p(\theta, s)$ and visualising it using θ and s as the axis. A single dot in the plane projected by the radon transformation will have a sinusoidal shape in the sinogram, due to the rotation of the scanning angle, which explains the origin of the name.

An example of a sinogram can be seen in figure 2.3 where a simulated image containing an circle and a square has been generated. The sinogram then arises by forward projecting the image using the Radon Transformation. Distinct sinusoidal bands with different thickness and intensities can be seen corresponding to the two shapes in the image.

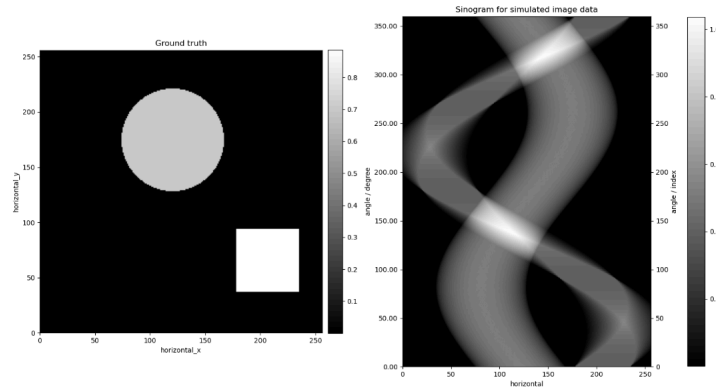


Figure 2.3: Example of sinogram made from a forward projection of simulated image data.

2.5 Computed Tomography

Computed tomography, also known as CT, is a computer-assisted imaging technique that typically uses X-rays to generate three-dimensional representations of both the exterior and interior of scanned objects. CT uses the attenuation coefficients of X-ray-penetrated objects to reconstruct the signal in an image. Looking at eq: (2.3) which describes the transmission of X-rays through non-homogeneous material. This is an exponential model, and it can thereby be more convenient to use a linear model for reconstruction. eq: (2.3) can be rewritten to:

$$-\log \frac{I}{I_0} = \int_L \mu(x) dx \quad (2.11)$$

$-\log \frac{I}{I_0}$ is called the absorption, and is a measurement for how many X-ray photons get absorbed by the material. A values of 0 means no material between source and detector. Generally, a CT setup consists of three essential components: an X-ray source, table or pedestal on which the object being tested is placed, and a detector system.

2.5.1 Scan Geometry

The Lambert-Beer law eq: (2.11) describes the integrated transmission values along a single thin X-ray line, and it would thereby be difficult to get any usable information about the material that the X-ray has travelled through with just a single measurement. It is therefore necessary to perform the scan from multiple directions around the object to gather useful information and be able to make a reconstructed image. Different geometries can be deployed when making a scan. The following geometries are usually used as geometries in 2D, but can be extended to 3D scanning by making multiple scans in the vertical dimension. The data used in this project will be constructed using this method.

2.5.1.1 Parallel-Beam Geometry

A parallel beam projection involves transmitting X-ray signals in a parallel manner, one after the other along a line at each angle. An illustration of this can be seen in figure 2.4a . The rays are attenuated as they pass through the object, and the corresponding attenuated signals are captured by the detector placed on the opposite side. Afterwards the detector and X-ray source (or the object) is rotated for a new set of parallel lines at a different projection angle. Using the parallel beam projection method in a laboratory set-up requires a lot of time for scanning each ray one by one and is thereby time-consuming method. In a synchrotron is the divergence of the beam small enough that parallel beam is used.

2.5.1.2 Fan-Beam Geometry

As a result of the parallel beam projection method being time-consuming, the Fan-beam projection method can be employed instead. Fan-beam involves placing a single X-ray source emitting a divergent beam and a curved or linear panel of detectors capturing the attenuated signal along the ray of paths on the other side of the object. An illustration can be seen in figure 2.4b. The source and detector panel (or the object) are then rotated to capture the signal from a different projection angle. By using this setup, the scanning time can be reduced because multiple transmission values can be captured at the same time.

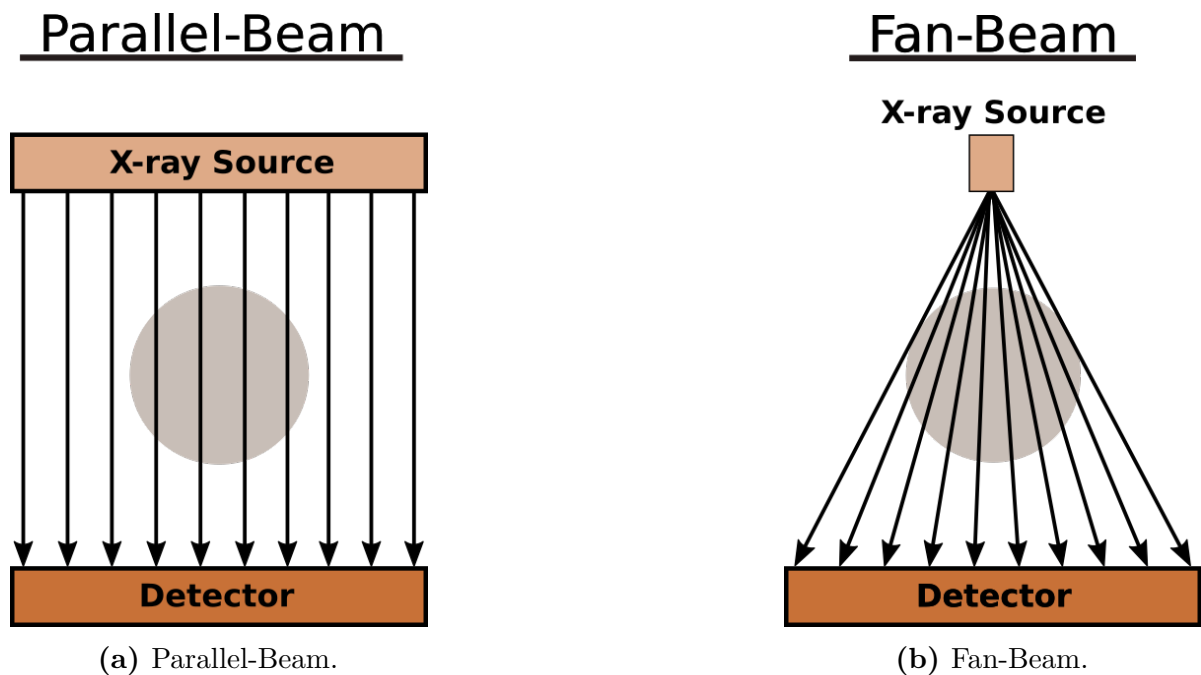


Figure 2.4: Illustration of different geometries.

2.6 Signal Quality

Different factors can cause the quality of the imaging signal to decrease.

2.6.1 Signal-to-Noise Ratio

In radiology, the signal-to-noise ratio (SNR) measures the ratio of true signal, which reflects the actual object of interest, to noise, such as random quantum noise. Since the aim is to reconstruct the object of interest any signal present in air detected is considered noise. The SNR is found as the difference between the signal and the background noise, divided by the standard deviation of the signal from the background, indicating the variability of the background noise. This can be formulated as:

$$SNR = \frac{\mu(phantom) - \mu(background)}{\sigma(background)}, \quad (2.12)$$

where μ and σ denote the mean value and the standard deviation. The SNR is closely related to the acquisition time, since longer acquisition time usually leads to less noise in the signal.

2.6.2 Noise

The primary source of noise focused on in this project is quantum noise, which results from the statistical nature of X-rays. The process of detecting photons follows a Poisson distribution, meaning that the noise amplitude is proportional to the square root of the signal amplitude, and the signal-to-noise ratio (SNR) also follows this relationship.

Using the same notation as in section 2.2 the intensities I_0 and I can be thought of as discrete numbers of the total energy of the X-ray photons incident on and transmitted through the object of interest. We can write the transmitted intensity I as a product of the incident transmission I_0 and the dampening as:

$$I = I_0 d \quad \text{with} \quad d = e^{-\int \mu(\ell) d\ell} \quad (2.13)$$

The probability that each photon will be detected is represented by d , and each photon is considered to be independent of all others. This means that the number of X-ray photons that are detected during a specific period can be considered a Poisson process.

2.6.2.1 The Poisson Distribution

The Poisson distribution is a statistical distribution which serves as an approximation for the number N occurrences of independent or close to independence event in a time

frame. The Poisson distribution is formulated by:

$$P(N = k) = \frac{\mu^k}{k!} e^{-\mu}, \quad (2.14)$$

where $P(N = k)$ denotes the probability that the number of events N occurring is k . μ is the intensity at which the events happens. An X-ray photon hitting the detector can be seen as Poisson distributed event. This event can be modelled as a random variable that follows a Poisson distribution with a parameter $I_0 d$, as formulated above:

$$I \sim \mathcal{P}(I_0 d), \quad (2.15)$$

where $\sim$ means that the variable I follows the Poisson distribution with intensity $I_0 d$. When $I_0 d$ increase a higher intensity signal is measured and the standard deviation of the noise also increases, but only with the square root of the intensity $\sqrt{I_0 d}$. This can be described in terms of signal to noise ration (SNR) with:

$$\text{SNR} = \frac{I_0 d}{\sqrt{I_0 d}} = \sqrt{I_0 d} \quad (2.16)$$

CHAPTER 3

Reconstruction Methods

This chapter will present the reconstruction method that is used throughout this project. The chapter will outline the setup used to go from data to a reconstructed image. The literature used for the chapter is: [3], [11], [12], [13], and [14]

3.1 Rays through Pixels

As described in section 2.5 the aim of computed tomography is to create a reconstructed image representing the scanned object of interest. The image will represent the object by displaying the attenuation coefficient of the material contained in pixels in the reconstructed image. The detector captures the X-ray intensity described by Lambert-Beer's law with eq: (2.2) from section 2.2:

$$I(l) = I_0 e^{-\int_L \mu(x) dl}$$

By the time the X-ray reaches the detector, it will have passed multiple image pixels as illustrated in figure 3.1. These are the pixels we wish to 'fill out' when making the reconstruction.

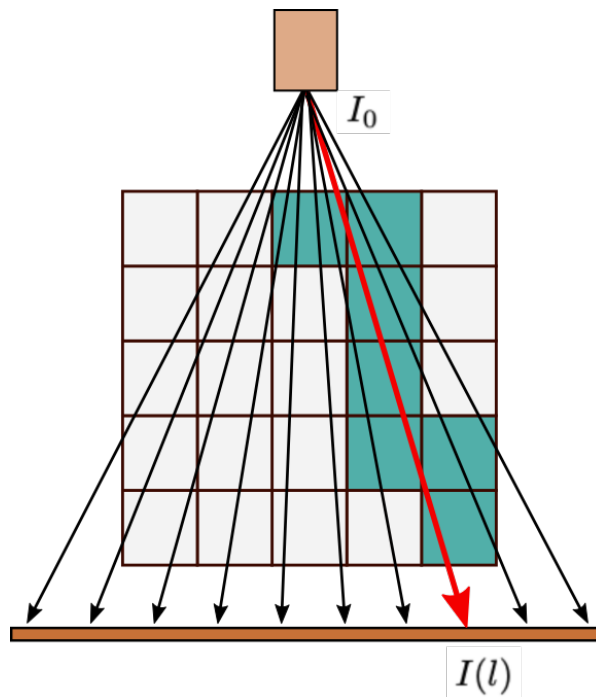


Figure 3.1: X-ray through object divided into pixels in the reconstructed image.

3.2 Setting up the System Matrix

A system of equations of the type $\mathbf{Ax} = \mathbf{b}$ can be formulated to describe the relation between the measured transmission data and the attenuation coefficients in the reconstructed image. The right hand side $\mathbf{b}$ is defined as the absorption:

$$b_i = -\log \frac{I_i}{I_0} = \int_{L_i} \mu(s) ds, \quad (3.1)$$

where L_i is the i -th ray. The number of elements in $\mathbf{b} \in \mathbb{R}^m$ is the number of lines in all views $m = N_s \cdot N_\theta$, where N_s is the number of detector pixels and N_θ is the number of projection angles. The system matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ consists of elements a_{ij} , representing the path length of the ray through the j -th pixel. The number of columns in $\mathbf{A}$ is the number of pixels in the image $n = \text{rows} \cdot \text{columns}$. Lastly a vector of unknowns $\mathbf{u} \in \mathbb{R}^n$ with elements u_j which is the j -th pixel value is defined. The system of equation then becomes:

$$\sum_{j=1}^n a_{ij} u_j = b_i \quad \text{For } i = 1, 2, \dots, m \quad (3.2)$$

or written in matrix notation:

$$\begin{bmatrix} a_{11} & \cdots & a_{1,j} & \cdots & a_{1,n} \\ \vdots & \ddots & & & \vdots \\ a_{i,1} & & a_{i,j} & & a_{i,n} \\ \vdots & & & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,j} & \cdots & a_{m,n} \end{bmatrix} \begin{bmatrix} u_1 \\ \vdots \\ u_j \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_i \\ \vdots \\ b_m \end{bmatrix} \quad (3.3)$$

3.2.1 Least-Squares Problem

An exact solution of the linear system is in practical not possible due to various issues. Problems may arise from the ill-posed nature of the matrix $\mathbf{A}$, its size being too big to make computing the inversion unfeasible, or the unavailability of an inverse for $\mathbf{A}$. Additionally, the presence of noise in the vector $\mathbf{b}$ can make an exact solution impossible. Instead an optimisation problem can be formulated by finding a solution such that $\mathbf{Au}$ is as close as possible to $\mathbf{b}$. The optimal solution $\mathbf{u}^*$ is then found by minimising a least-squares fit:

$$\mathbf{u}^* = \arg \min_{\mathbf{u}} \|\mathbf{Au} - \mathbf{b}\|_2^2 = \sum_{i=1}^n ((\mathbf{Au})_i - b_i)^2 \quad (3.4)$$

3.3 Iterative Reconstruction Methods

The least-squares optimisation problem presented in eq: (3.4) can be solved using iterative reconstruction methods. The workflow for iterative reconstruction methods consist of three major steps which are repeated until a solution is found. The steps are visualised in figure 3.2. The process usually starts with an empty image or an image containing some prior information. The initial image is forward projected generating artificial raw data. Subsequently, these artificial raw data are compared to the actual measured raw data, leading to the computation of a correction term. In the final step, the correction term is back-projected onto the volumetric object estimate, to generate a new initial image. The iterative process terminates either after a fixed number of iterations, when the update for the current image estimate is considered sufficiently small, or when a predefined quality criterion in the image estimate is met.

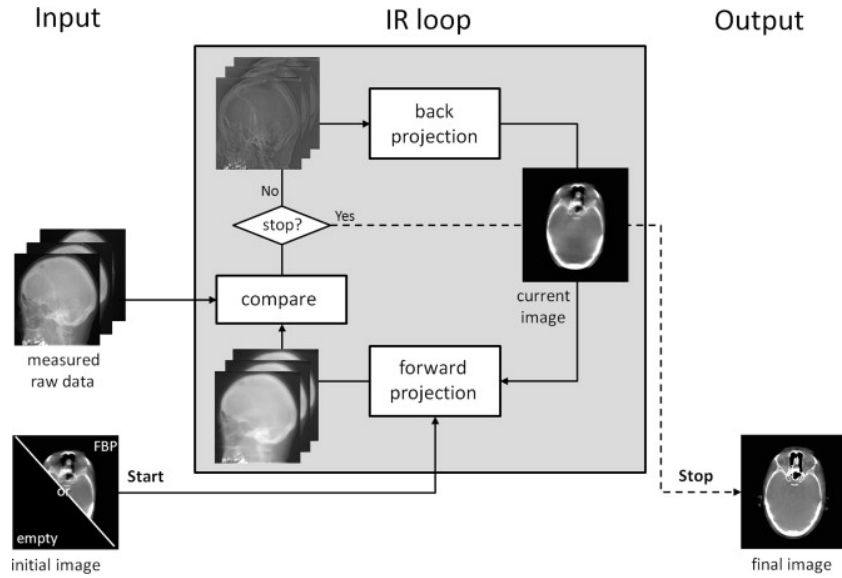


Figure 3.2: Workflow for iterative reconstruction. Source [11].

3.3.1 Simultaneous Iterative Reconstruction Technique (SIRT)

The Simultaneous Iterative Reconstruction Technique (SIRT) is an iterative reconstruction technique used to solve a weighted version of the optimisation problem seen in eq: (3.4). The method is of the type: Fully Simultaneous Method, and therefore utilise all the rows of $\mathbf{A}$ simultaneously in one iteration. The algorithm uses the two diagonal matrices $\mathbf{R} \in \mathbb{R}^{m \times m}$ and $\mathbf{C} \in \mathbb{R}^{n \times n}$ where m and n are the same as described in section 3.2; which is m as the number of rows in $\mathbf{A}$ and n as the number of columns in $\mathbf{A}$.

The diagonal elements of $\mathbf{R}$ are the inverse row sums found as:

$$r_{ii} = \frac{1}{\sum_j (a_{ij})} \quad (3.5)$$

The diagonal elements of $\mathbf{C}$ are the inverse column sums found as:

$$c_{jj} = \frac{1}{\sum_i (a_{ij})} \quad (3.6)$$

For both $\mathbf{R}$ and $\mathbf{C}$ are elements not in the diagonal set to 0. The algorithm have the option to include a non-negative constraint such that $u_i \geq 0$. This is to accommodate that fact that attenuation values are positive values. The simple constraint to incorporate this is $\mathbf{u} = \max(0, \mathbf{u})$.

3.3.2 SIRT Algorithm

The SIRT algorithm to solve our optimisation problem presented is then:

Algorithm 1: SIRT

Input: $\mathbf{A}$, $\mathbf{u}_{[0]}$, $\mathbf{b}$, α , S : as the number of iterations

Output: $\mathbf{u}_{[S]}$

Set $\mathbf{R} = \text{diag} \frac{1}{\sum_j (a_{ij})}$;

Set $\mathbf{C} = \text{diag} \frac{1}{\sum_i (a_{ij})}$;

for $s=1, 2, \dots, S$ **do**

$\mathbf{u}_{[s]} = \mathbf{u}_{[s-1]} + \mathbf{CA}^\top \mathbf{R}(\mathbf{b} - \mathbf{A}\mathbf{u}_{[s-1]})$

end

3.4 Regularisation

The least-squares problem can be ill-posed and small changes to the right-hand side $\mathbf{b}$ can lead to large changes in the least-squares solution, $\mathbf{u}$. Noise in the raw data will therefore lead to large errors in the least-squares solution. To avoid this regularisation will be used in the iterative reconstruction. A Least-squares optimising problem with regularisation takes the form:

$$\mathbf{u}^* = \arg \min_{\mathbf{u}} \|\mathbf{A}\mathbf{u} - \mathbf{b}\|_2^2 + \alpha \cdot J(u), \quad (3.7)$$

where $J(u)$ is called the regularisation term, and α is a positive regularisation parameter that controls the weighting between the two terms in the objective function eq: (3.7). A large regularisation parameter will make it so the solution is almost only affected by the regularisation, while a small regularisation parameter will make it so the solution is close to finding the least-squares solution. Different regularisation terms will have different influences on the solution. This project will primarily use Total Variation regularisation (TV).

3.4.1 Tikhonov Regularisation

Tikhonov regularisation is one of the most widely used regularisation techniques. The idea behind Tikhonov is to use the norm of $\mathbf{u}$ to control the dominant values, with large amplitudes, in the solution to the otherwise non-regularised least-squares problem (eq: (3.4)). Tikhonov regularisation is formulated as:

$$\mathbf{u}^* = \arg \min_{\mathbf{u}} \|\mathbf{A}\mathbf{u} - \mathbf{b}\|_2^2 + \alpha \cdot \frac{1}{2} \|\mathbf{L}\mathbf{u}\|_2^2 \quad (3.8)$$

$\mathbf{L}$ is a matrix that can be suitably chosen to fit the problem.

In this project will $\mathbf{L} = \mathbf{I}$ be used such that the regularisation problem becomes:

$$\mathbf{u}^* = \arg \min_{\mathbf{u}} \|\mathbf{A}\mathbf{u} - \mathbf{b}\|_2^2 + \alpha \cdot \|\mathbf{u}\|_2^2 \quad (3.9)$$

We thereby regularise our solution towards a solution with a lower 2-norm. This is also know as $\mathbf{L}_2$ -regularisation.

3.4.2 Total Variation regularisation (TV)

The Total Variation regularisation (TV) is designed to enhance edges in the image. TV is based on the observation that images with sharp edges or transitions have high variation in pixel intensities compared to smooth regions. TV regularisation therefore encourages the reconstruction algorithm to preserve sharp edges and fine details while smoothing out noise. Since this project aims to emulate baggage scanning data, where the materials are homogeneous and have sharp edges TV regularisation have been chosen. In this project the TV optimisation problem will be defined as:

$$\mathbf{u}^* = \arg \min_{\mathbf{u}} \|\mathbf{A}\mathbf{u} - \mathbf{b}\|_2^2 + \alpha \cdot \text{TV}(u) \quad (3.10)$$

$$= \arg \min_{\mathbf{u}} \|\mathbf{A}\mathbf{u} - \mathbf{b}\|_2^2 + \alpha \cdot \|\mathbf{D}\mathbf{u}\|_{2,1}, \quad (3.11)$$

where $\mathbf{D}$ represents the finite difference approximation to a gradient operator. In a 1D signal represented by $x \in \mathbb{R}^p$ is the matrix $\mathbf{D}$:

$$\mathbf{D}_P = \begin{pmatrix} -1 & 1 & & & \\ & \ddots & \ddots & & \\ & & -1 & 1 & \\ & & & & 0 \end{pmatrix} \in \mathbb{R}^{P \times P} \quad (3.12)$$

In a 2D CT image of size $M \times N$ the vector $u \in \mathbb{R}^n$ represent the image by stacking its pixels in a row and setting $n = N \cdot M$. The approximation of the gradient operator $\mathbf{D}_n$ in 2D is then:

$$\mathbf{D}_{M \times N} = \begin{pmatrix} \mathbf{D}_x \\ \mathbf{D}_y \end{pmatrix} \in \mathbb{R}^{2n \times n}, \quad (3.13)$$

where $\mathbf{D}_x$ and $\mathbf{D}_y$ is the finite different in the x and y direction respectively. The isotopic TV regularisation term which will be used in this project is:

$$\text{TV}(\mathbf{u}) = \|\mathbf{D}\mathbf{u}\|_{2,1} \quad (3.14)$$

$$= \sum_{j=1}^n \sqrt{[(\mathbf{D}_x)\mathbf{u}]_j^2 + [(\mathbf{D}_y)\mathbf{u}]_j^2} \quad (3.15)$$

3.4.3 Fast Iterative Shrinkage Thresholding Algorithm (FISTA)

Since Total Variation regularisation is not a smooth function and therefore not differentiable, the optimisation problem defined in eq: (3.11) will be solved using the Fast Iterative Shrinkage Thresholding Algorithm (FISTA). FISTA is primarily designed for solving convex optimisation problems of the form:

$$\arg \min_x \{f(x) + g(x)\}, \quad (3.16)$$

where $f(x)$ is a convex function and $g(x)$ represents a possible non-smooth regularisation term. $f(x)$ is continuously differentiable with Lipschitz constant $L(f) > 0$

3.4.3.1 Proximity Mapping

FISTA uses proximity mapping to determine the correction term in the algorithm. A quadratic approximation of the function $f(x) + g(x)$ at a given point $\mathbf{y}$ is:

$$Q_L(\mathbf{x}, \mathbf{y}) := f(\mathbf{y}) + \langle \mathbf{x} - \mathbf{y}, \nabla f(\mathbf{y}) \rangle + \frac{L}{2} \|\mathbf{x} - \mathbf{y}\|^2 + g(\mathbf{x}) \quad (3.17)$$

which will have an a unique minimiser:

$$p_L(\mathbf{y}) := \operatorname{argmin} \{Q_L(\mathbf{x}, \mathbf{y}) : \mathbf{x} \in \mathbb{R}^n\}. \quad (3.18)$$

This can be expressed as:

$$p_L(\mathbf{y}) = \operatorname{argmin}_{\mathbf{x}} \left\{ g(\mathbf{x}) + \frac{L}{2} \left\| \mathbf{x} - \left(\mathbf{y} - \frac{1}{L} \nabla f(\mathbf{y}) \right) \right\|^2 \right\}. \quad (3.19)$$

where $\left(\mathbf{y} - \frac{1}{L} \nabla f(\mathbf{y}) \right)$ works as a step with gradient descent.

3.4.3.2 FISTA Algorithm

The FISTA algorithm for our optimisation problem presented in eq 3.11 is:

Algorithm 2: FISTA with constant step size.

Input: $L(f)$: as the Lipschitz constant, $\mathbf{u}_{[0]}, \mathbf{b}, \alpha$, S : as the number of iterations

Output: $\mathbf{u}_{[S]}$

Set $\mathbf{y}_{[1]} = \mathbf{u}_{[0]}$;

Set $t_{[1]} = 1$;

for $s=1, 2, \dots, S$ **do**

$\mathbf{u}_{[s]} = p_L(\mathbf{y}_{[s]});$

$t_{[s+1]} = \frac{1 + \sqrt{1 + 4t_{[k]}^2}}{2};$

$\mathbf{y}_{[k+1]} = \mathbf{u}_{[k]} + \left(\frac{t_{[k-1]}}{t_{[k+1]}} \right) (\mathbf{u}_{[k]} - \mathbf{u}_{[k-1]})$

end

CHAPTER 4

Data Processing Techniques

4.1 Data Correction

The detector captures counts the number of photons in each energy bin, and the raw data files therefore contains finite countable data, known as intensity. Photon counting detectors can be subject to various spectral distortions, where interactions between photons can results in a wrong energy detection and/or activation of neighbouring photons. It is therefore necessary to use a data correction algorithm on the collected detector data before sinograms can be constructed. The algorithm addresses disturbances such as charge sharing, fluorescence radiation, scattering radiation, and pulse pile up. The full explanation of these distortions can be seen the work [15]. Given that this project utilises an existing Matlab script for the correction process, further elaboration on the method is therefore not outlined. The algorithm used was developed in [16].

Before the data can be transformed into absorption, the transmission data will have to be corrected using a flat field to corrugate for background noise. A flat field image (I_F) is captured, which is an image with the source on, but without an object between the source and the detector. The measured transmission images (I) is then flat field corrected by the flat field resulting in the projection data (I_p) using:

$$I_p = -\log\left(\frac{I}{I_F}\right), \quad (4.1)$$

where the division between I and I_F is element wise division. In our experimental set-up is the flat field captured after the full scan full scan to guarantee that the X-ray source is operation at full potential. After this correction process does the data represent the absorption.

4.2 Removing Dead/Hot Pixels

The detector can sometimes have dead pixels, which are pixels where no signal is detected from the X-ray source. This will lead to dark pixels in the sinogram, and appears as lines when looking over all the projections. Dead pixels will need to be removed to not disturb the reconstruction. The removal can be done by creating a mask, which is a binary image containing 1 or 0, where 0 notates pixels that are undesirable. The mask will be generated using the `MaskGenerator.mean` from the Core Image Library (CIL) presented in [17]. The function generates a mask for values outside a multiple of the

standard-deviation from the mean:

$$\text{abs}(I - \text{mean}(I)) < \tau \cdot \text{std}(I), \quad (4.2)$$

where τ is a threshold factor that dictates how close to the standard-deviation from the mean 'good' pixels have to be. The pixels identified in the mask as being outliers are then interpolated using the CIL function `Masker.interpolate` [17] with the nearest neighbour method, to find the pixels desired values.

4.3 Detector Gap Interpolation

The detector used for the experimental portion of this project is a detector consisting of multiple detector modules which can be daisy chained together to form a detector array of various pixel length. This results in small gaps between the detector modules, The photons landing in these gaps are therefore not detected, and the captured data will be missing some data points. The size of the gaps between the detector modules will be measured during scanning and will be documented in centimetres. The processing is done in CIL and the gap will be approximated to the nearest integer to fit in order to meet the requirement of the gap being in whole pixels. The amount of pixels corresponding to the detector gap will be inserted at every projection along the horizontal axis. The sinograms, and consequently the data set, will thereby become larger in the horizontal dimension of the possessed data compared to the originally acquired data. The values of the inserted pixels will be found using linear interpolation one angle at a time by only using the known pixel values from that angle.

4.4 Ring Remover

The sinogram occasionally have unwanted horizontal stripes that will show on the reconstruction as ring artefacts. In the event that these rings are excessively prominent, they will be dampened from the sinogram using the function `RingRemover` from CIL. The function and source code are based on the paper [18]. The paper introduces a filtering technique that firstly uses a wavelet filter to decompose the image into separate bands of horizontal, vertical, and diagonal details at various resolution scales. The bands undergo Fast Fourier Transformation (FFT) to further narrow down the stripe details into compact bands. The condensed stripe information is then suppressed by multiplying it with a Gaussian damping function. Finally, the filtered coefficients are used to reconstruct an image without stripes. An example of the effect of `RingRemover` can be seen in figure 4.1, where artificial noise have been added to an sinogram, and then removed again using the function.

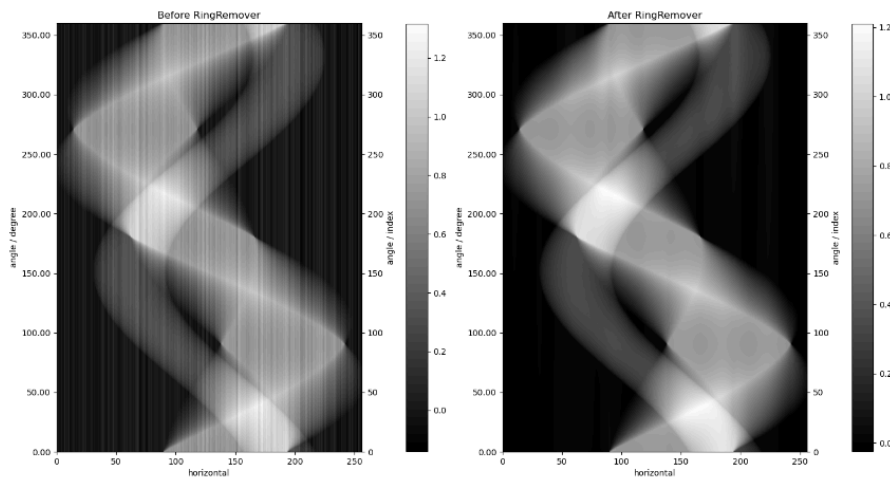


Figure 4.1: Example of the use of `RingRemover`.

CHAPTER 5

Figure of Merit

In order to assess the quality of a reconstruction, a figure of merit (FOM) needs to be specified. Different FOMs will be proposed, utilised, and evaluated.

5.1 Figure of Merit Based on a Reconstructed Image

The first FOM proposed makes use of a ground truth image to compare reconstructions against. The root mean squared error (RMSE) will be used to compare the two images:

$$\text{RMSE}(I, G) = \sqrt{\frac{1}{m \cdot n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - G(i, j)]^2}, \quad (5.1)$$

where m is number of horizontal pixels and n is number of vertical pixels. I is the reconstructed image and G is the ground truth image.

In order to use this FOM it is essential to establish a ground truth image, which serves as a reference for assessing the quality of the reconstructions. Real data is used for this project and a ground truth image will therefore not be perfect, since there is no simulated data which has been forward projected. As a result of this will for each data set employed in this project, distinct ground truth images will be created and carefully selected. The construction of the ground truth reconstruction will utilise all available data within a given data set. Consequently, the quality of the ground truth image may vary from one data set to another. In the reconstruction each energy channel will be reconstructed individually without influence from each other. Thus, the number of channels employed for the reconstruction should not impact the outcome. For each data set employed in this project, distinct ground truth images will be created and carefully selected.

5.2 Figure of Merit Based on Energy Curves

A second FOM proposed is a metric based on attenuation curves extracted from materials in the reconstruction. Materials in baggage scanning are classified based on their attenuation coefficients, and it is therefore important that the reconstruction results in the right attenuation values for the materials.

5.2.1 Finding Ground Truth Curves

Table values for attenuation coefficients can easily be found in databases such as the NIST XCOM. An Example of NIST XCOM curves for 5 materials (Aluminium, Dichloromethane, PVC, Ethanol, and Water) can be seen in figure 5.1. The curves have been found using the `xraydb` python library [19]. While table values can be useful to get a basic understanding on how the attenuation coefficient should behave, can they also be misleading due to local deviations in the x-ray set up. For example can diverse materials, such as PVC be combined with other materials from sample to sample which complicates the calculation of table values. It is therefore important to generate a comprehensive collection of ground truth attenuation curves directly derived from the actual data employed. The table value curves seen in 5.1 will be used to evaluate the found ground truth curves from a visual perspective, but the figure of merit will be based on the curves created from the real data. Due to large scanning uncertainties in low energies channels and very similar attenuation for the materials in high energy channels, the error will always be evaluated in the energy span: $[40.151 - 103.14]$ keV, which is the span seen in figure 5.1b.

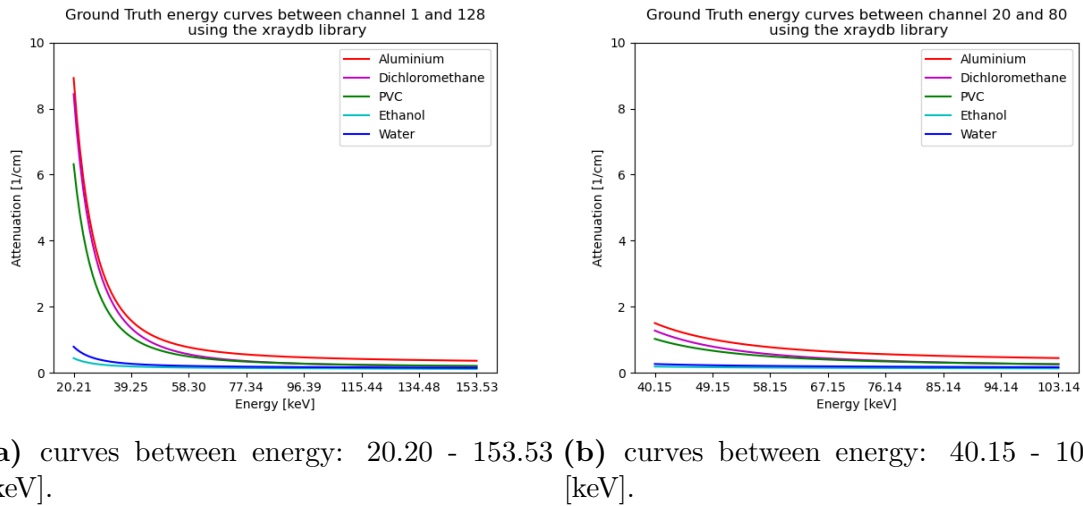


Figure 5.1: xraydb energy curves.

5.2.2 Pixels Sampling

The energy curves will be constructed by sampling a number of pixels from the materials scanned in each channel. The curves can be constructed using two different sampling methods:

- **Random pixels:** The method samples 100 random pixels in each material.
- **Small area:** The method sample a small area of 20×20 pixels in the middle of each material.

The sampled pixels are then averaged to find 1 attenuation coefficient for each energy channel in each material.

5.2.3 Error Calculation

The set of ground truth energy curves are then used to find the error in reconstruction. The error between ground truth curves and experimental curves are evaluated using two different methods:

- **Mean $\rightarrow$ Error:** After sampling the pixel values are averaged before finding the distance to the ground truth:

$$Error(channel = i) = \left| \left(\sum_{pixels} \frac{I_i}{N} \right) - G_i \right| \quad (5.2)$$

- **Error $\rightarrow$ Mean:** After sampling the distances to the ground truth are found, and the values are then averaged:

$$Error(channel = i) = \sqrt{\sum_{pixels} \frac{1}{N} [I_i - G_i]^2} \quad (5.3)$$

I_i is sampled values in channel i , G_i is the ground truth value in channel i and N is the number of pixels sampled.

With the two different method for sampling the pixels and two different method for calculating the error, 4 different energy curve based figure of merit can be defined. The different methods and the notation for them throughout the rapport can be seen in table 5.1

Table 5.1: 4 Method for finding the error using different sampling methods and error finding method.

Sampling \ Method	Mean $\rightarrow$ Error	Error $\rightarrow$ Mean
Random Pixels	Small Area, Mean	Small Area, Error
Small Area	Random Pixels, Mean	Random Pixels, Error

CHAPTER 6

Preliminary Study

A number of preliminary studies will be performed on a small already existing data set to examine reconstruction parameters and figure of merits.

6.1 Data

The X-ray data used are a multi spectral scanning of 5 materials, which are; Powdered Sugar, H_2O , H_2O_2 , 10 mm aluminum, and 7.8 mm PVC. The 3 materials; Powdered Sugar, H_2O , and H_2O_2 are stored in small glass containers. The placement of the materials in the set-up can be seen in figure 6.3b. The dimensions of the data and number of projections are stated in table 6.1. It can be seen from table 6.1 that the data only has 1 vertical slice and can thereby be processed as a 3D data set by 'collapsing' the vertical dimension. The geometry used is a 2D fan beam and the acquisition parameters for the data can be seen in table 6.2. An illustration of the geometry can be seen in figure 6.1.

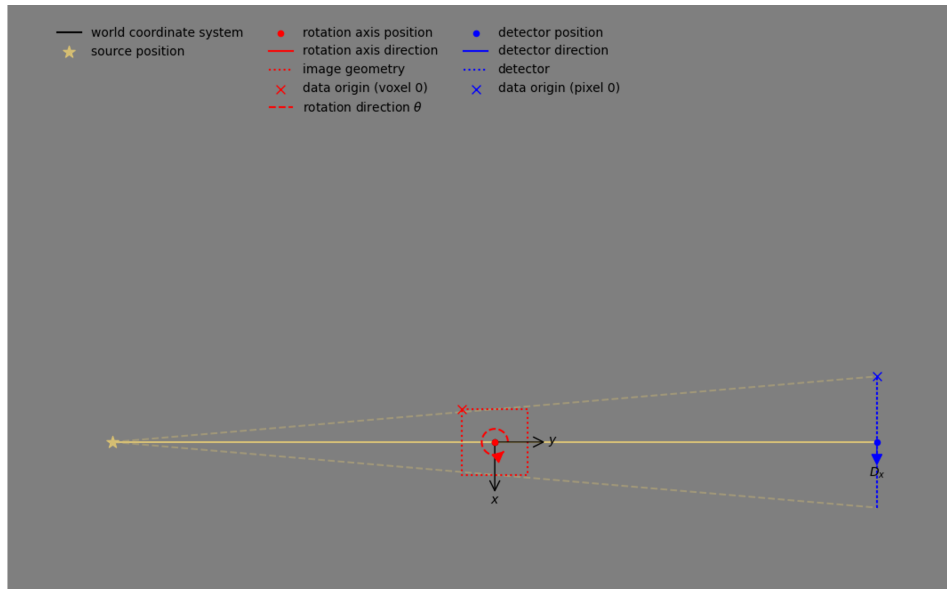


Figure 6.1: Geometry set up for preliminary data set.

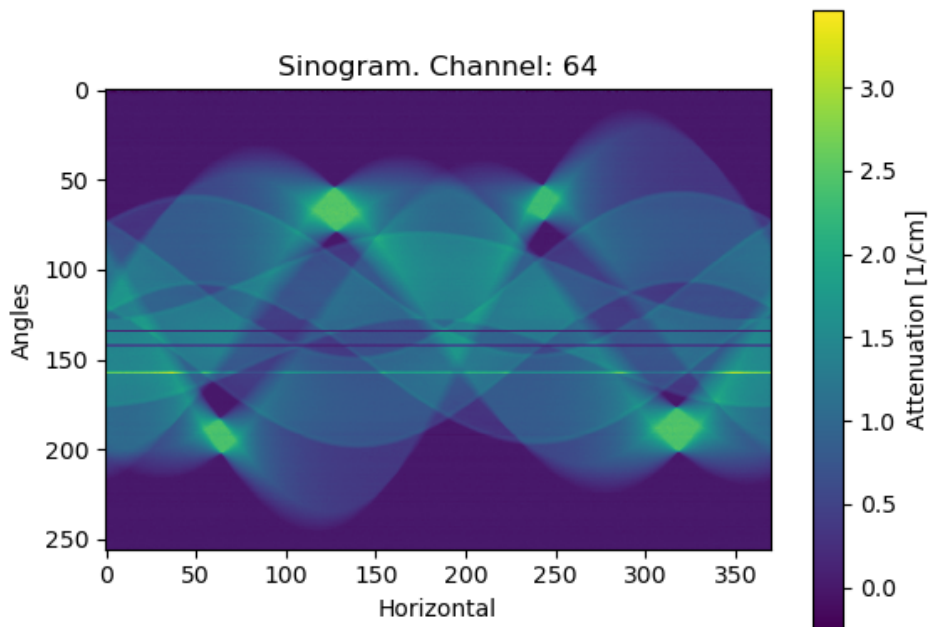
Table 6.1: Dimension of data used

Projections	Vertical slice	Horizontal slice	Energy channels
370	1	256	128

Table 6.2: Acquisition parameters

Projections	Angular span	Source Detector distance	Source - - Axis Of Rotation distance
370	360 Degrees	115 cm	57.5 cm

The data used has already been corrected from the raw data files using the algorithm described in 4.1 to form a sinogram for each energy dimension. Figure 6.2 shows the sinogram for channel 64 at 86.34 keV.

**Figure 6.2:** Sinogram for channel 64.

6.1.1 Experiment equipment

The detector used is the Multix ME100 photon counting detector with 128 energy channels [20]. The detector consists of 2 modules which have been put together to form a detector array and the modules will have a gap between them as outlined in 4.3. The detector parameters can be seen in table 6.3.

Table 6.3: Detector parameters.

Detector pixels (Excluding gap)	Detector modules	Pixel size	Detector gap	Energy span
256	2	0.077 cm	0.153 cm	[20.21 ; 153.53] keV

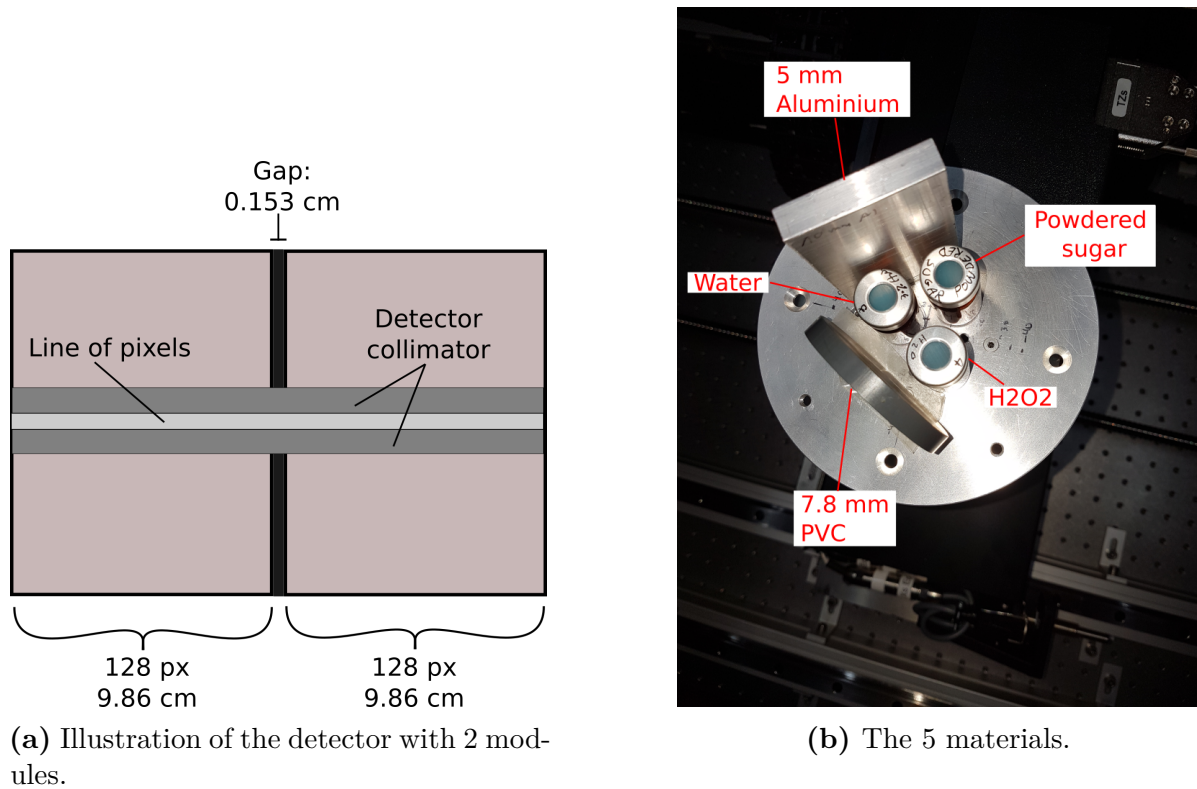


Figure 6.3: Detector and material setup.

An illustration of the detector with 2 modules used can be seen in figure 6.3a. The X-ray tube used is the Hamamatsu Microfocus X-ray source L12161-07 with a tungsten anode. The source is operated at a bias of 150 kV and filament current of 0.5 mA. Due to the use of a tungsten anode, the radiation spectra will have characteristic spikes such as the ones shown in figure 2.1, at approximately 60 keV.

6.1.2 Acquisition Repetitions

When the data was generated, a total of 10 repetitions each at 100 ms, were made for each projection, for a total capture time of 1 second pr. projection. This results in 10 times as large data set. Upon loading and correcting the raw files obtained from the detector using the algorithm from section 4.1, an optional number of repetitions can be used, which are then averaged condensing them into a single projection each angle. Having multiple repetitions should make the data less prone to noise, but the trade-off is longer capture time.

The effect on the acquired photons using multiple repetitions can be seen in figure 6.4, where the variance in photon counts for the 10 repetitions have been plotted against the average count. It can be seen from the figure that the number of acquired photons varies between repetitions. This variance can be smoothed out by using multiple repetitions.

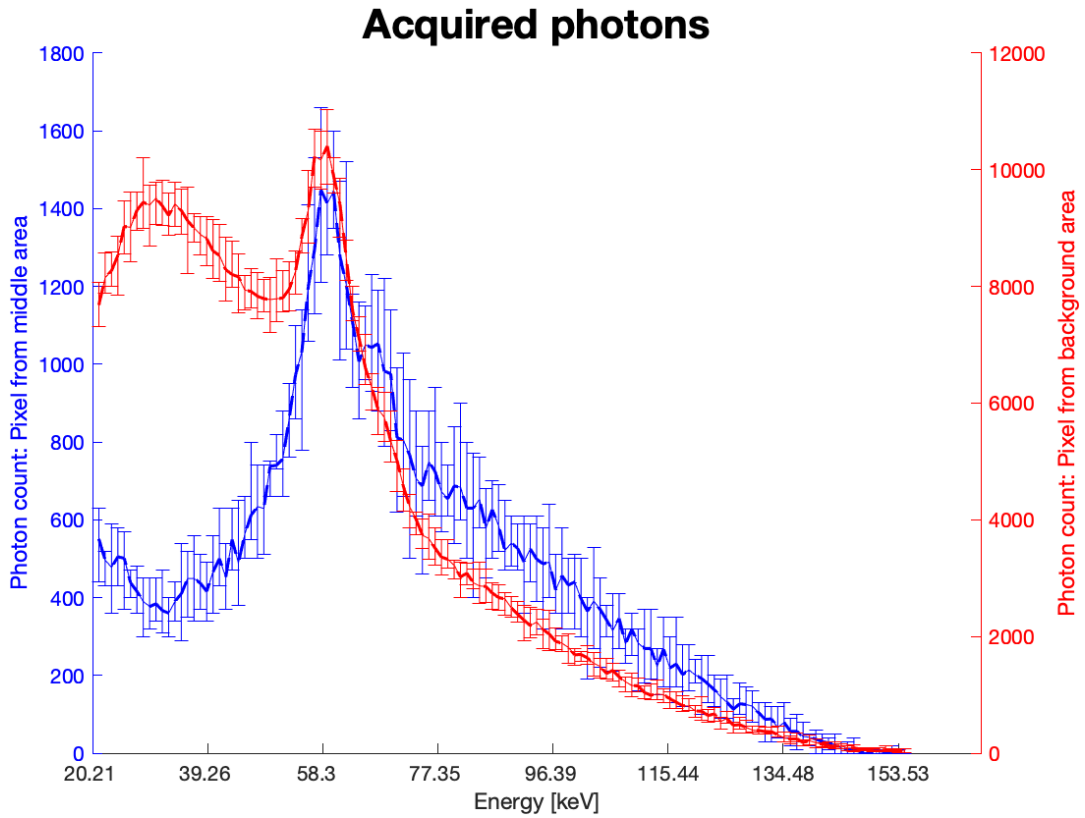
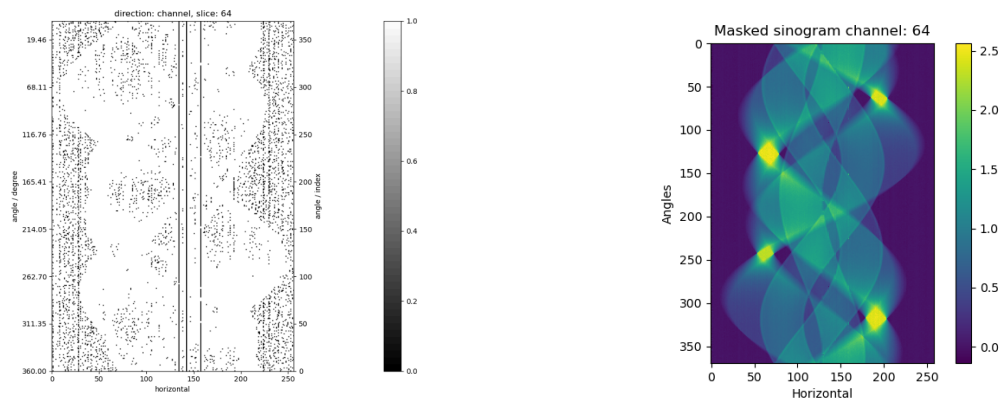


Figure 6.4: Acquired photons for a pixel in the background and in the middle of the scanned area. The graphs show the average of 10 repetitions, and the error bars shows the variation of the individually repetitions. Note that the two graphs have different scaling.

6.2 Preprocessing of Data

6.2.1 Removing dead/hot pixels

Looking at figure 6.2 it is seen that the acquired data has some unwanted dark and light pixels at all angles, which appears on the sinogram as lines. This is dead/hot pixels and needs to be addressed in order to avoid a degradation in the reconstruction due to this. The pixels will be removed using the method described in section 4.2. A threshold factor of $\tau = 1.5$ was employed and a window of 5 pixels was used to calculate the rolling mean. A mask is generated for each channel and the mask for channel 64 can be seen in figure 6.5a.



(a) Generated mask for sinogram using `MaskGenerator.mean` from CIL. (b) Sinogram after masking using `Masker.interpolate` from CIL.

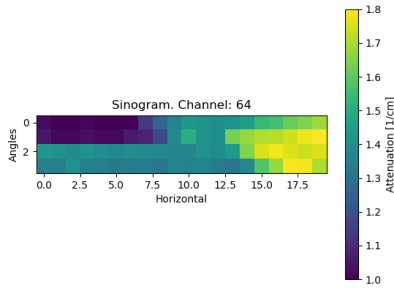
Figure 6.5: Mask used to remove dead pixels and sinogram after masking.

It can clearly be seen that the mask have identified the lines emerged from the dead/hot pixels. It is also important to note when looking at the mask (figure 6.5a) that `MaskGenerator.mean` method also identifies pixels that do not fall within the dark or bright lines such as pixels along the edge of the sinogram. This has been overlooked due to the transmission values in this area being very close to homogeneous and the interpolation should exhibit minimal deviation from the original values, since the nearest neighbor method has been used for the interpolation. Figure 6.5b shows the sinogram after using the mask and interpolation, and the faulty pixels are gone.

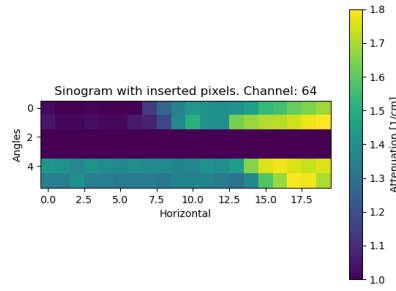
6.2.2 Dealing with the Detector Gap

As mentioned in section 6.1.1 the detectors used for capturing the data has two detector-panels each at 128 pixels with small gaps in between. The size of the gap is measured as 0.153 cm and will be interpolated using the method described in section 4.3. The method is restricted to whole pixels and therefore 2 pixels will be inserted in the sinogram since the pixel size is 0.077 cm.

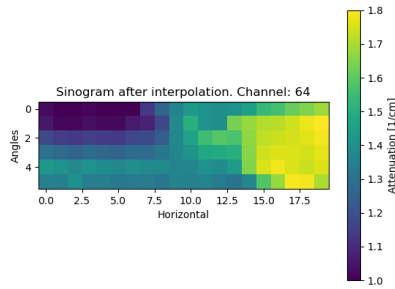
Figure 6.6a shows the sinogram zoomed in at horizontal pixels $[127 - 130]$ before pre-processing. It is easy to see that the transition from the two halves of the sinogram is very abrupt and some data is missing, which is the reason for inserting new pixels and interpolation to decide their values. Figure 6.6b shows how the sinogram looks with inserted pixels, which is initialised as 0. Figure 6.6c shows a zoomed in sinogram after interpolation. It can now be seen that the transition looks more smooth compared to that of before the inserting and interpolation of new pixels seen in figure 6.6a.



(a) Sinogram before interpolation.



(b) Insert of new pixels in sinogram.



(c) Sinogram after interpolation.

Figure 6.6: Zoom of sinogram during the detector gap processing. Angle: 0 to 20 and horizontal pixel: 126 to 130.

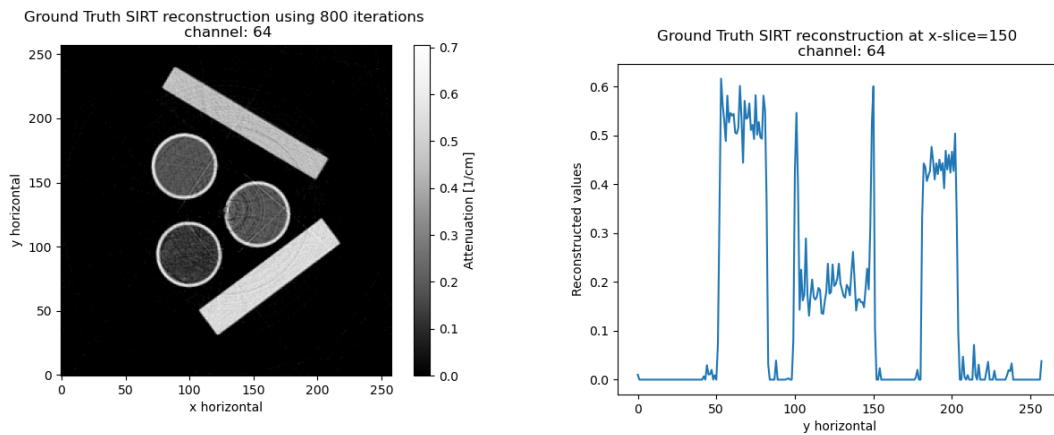
6.3 Ground Truth Reconstruction

We want to find a ground truth image in order to use the RMSE figure of merit presented in section 5.1. All available data will be used in the making of the ground truth; 370 projections, 128 energy channels, and 10 acquisition repetitions as described in section 6.1.2. Every energy channel will be reconstructed one-by-one without the influence of each other. Two ground truth reconstructions will be proposed.

6.3.1 Reconstruction with SIRT

The first proposed ground truth is a SIRT reconstruction using the method describe in section 3.3.1 with the non-negativity constraint. The reconstruction will run for 800 iterations to make sure that the algorithm is converged, since SIRT usually requires many iterations.

The resulting reconstructed image and a profile plot, which has been made by taking a slice at horizontal $x = 150$ seen in 6.7. The reconstruction looks sharp and the edges are not blurry. This can be seen by looking at the profile plot in figure (6.7b) that the edges for the glass container, seen in the middle of the plot, appears clear. The reconstruction do have some big ring artefacts which can be seen clearly in the two blocks in the reconstruction, and the profile plot shows a lot of noise.



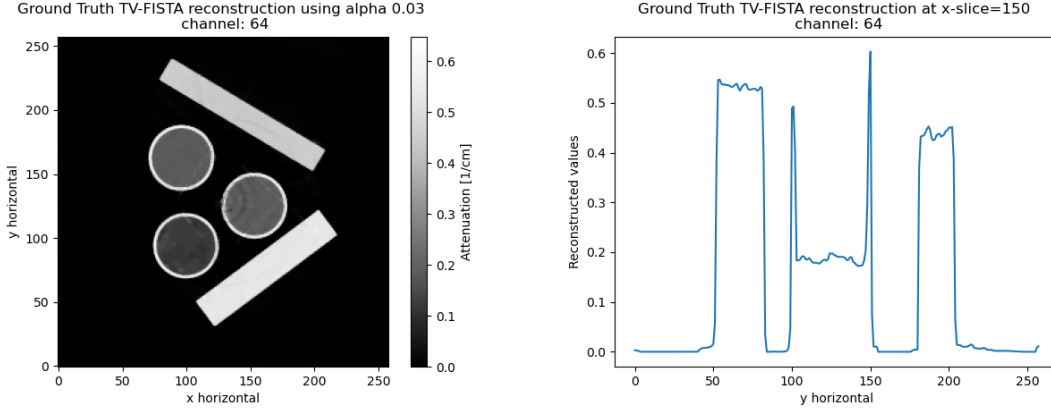
(a) Reconstruction using SIRT and 800 iterations. Channel 64. (b) Profile plot for the reconstructed image at $x=150$.

Figure 6.7: SIRT reconstruction with 800 iterations.

6.3.2 TV Reconstruction with Small Regularisation

The second proposed ground truth is a reconstruction using TV regularisation and FISTA as described in 3.4.2. It is important when using regularisation to choose the regularisation parameter correct, especially when creating a ground truth, to avoid bias in future reconstructions. The parameter has to be low enough that it does not affect the edges in the reconstruction, but still big enough that it has an effect on the noise in the reconstruction. Different parameters have been tested and the parameter of $\alpha = 0.03$ has been chosen as optimal. The reconstruction and a profile plot can be seen i figure 6.8. The profile is less noisy compared to the SIRT profile plot. The edges of the glass container seen in the middle of the plot is still sharp and has not been regularised too much, which would make the peaks smaller and closer to the material in the class. It can also be seen that the aluminum and PVC blocks in the sides of the profile plot

are more flat across the material, which they should be since they are homogeneous materials. The reconstruction itself also looks reasonable with no prominent artefacts. The ground truth used for the figure of merit will therefore be the reconstruction using TV regularisation with $\alpha = 0.03$ and solved using FISTA. The use of the `RingRemover` algorithm described in section 4.4 was not found to be necessary in the making of the ground truth image.



(a) Reconstruction using TV-FISTA and $\alpha = 0.03$. Channel 64. (b) Profile plot for the reconstructed image at $x=150$.

Figure 6.8: TV-FISTA reconstruction with $\alpha: 0.03$.

6.4 Choosing of an Optimal Regularisation Parameter

The choice of the regularisation parameter α is important to ensure a good but also precise reconstruction. An investigation on how to choose the parameter therefore needs to be performed. The optimal parameter value will be affected by the number of projections and signal quality. 6 different number of projections extracted from the data set will be used to study the regularisation parameter. The data set has 370 projections, and to ensure the integrity and avoid the exclusion of any projections, the downsampling of the data is exclusively done to factors of 370. Thereby the different numbers of projections used in this study are as follows:

$$5, 10, 37, 74, 185, 370 \quad (6.1)$$

6.4.1 Constant Photon Count

Various acquisition repetitions can be used in the reconstruction process as outlined in section 6.1.2. The photon count is kept constant, ensuring that for each down-

sampling of the data, the product of repetitions and projections equals a constant; repetitions $\cdot$ projections = K , where K should be the same for all configurations. This is done to emulate that baggage scanning have a fixed photon budget. As a result will a decrease in the number of projections requires more repetitions as seen in table 6.4, which outlines the projections/repetitions configurations tested.

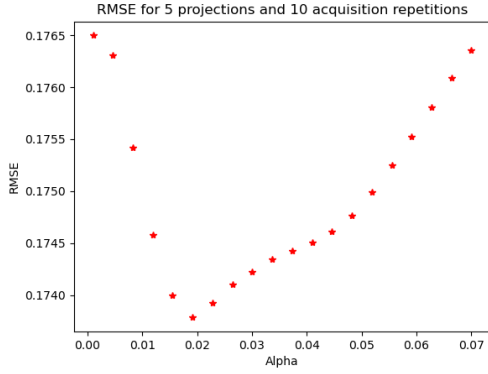
Table 6.4: Configurations tested.

Projections	5	10	37	74	185	370
Repetitions	10	10	10	5	2	1
Product	50	100	370	370	370	370

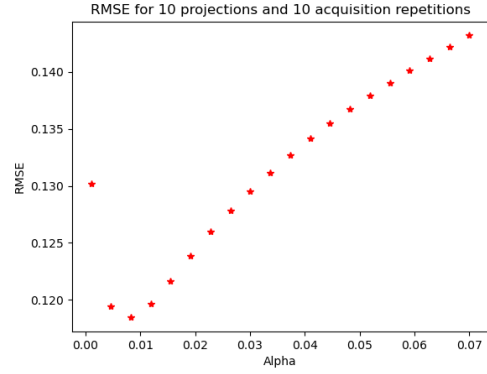
In table 6.4 10 repetitions have been used multiple times. This is due to the data set only having 10 repetitions and can not be downsized more. For each configuration of projections and repetitions 20 different reconstructions were made with 20 equally distant α values in the span:

$$\alpha = [0.001, 0.07] \quad (6.2)$$

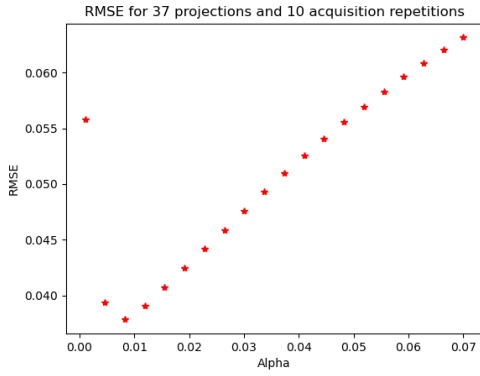
200 iterations of FISTA was used to make sure that the solution is converged. The reconstructions were then compare with the ground truth reconstruction described in section 6.3.2 using the RMSE from eq: 5.1 at channel 30. At this energy level there is enough signal to justify that the reconstruction is comparable across different reconstructions. The RMSE for each configuration across the α span are shown in figure 6.9.



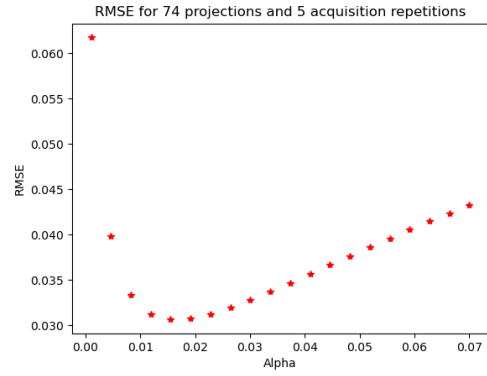
(a) 5 projections and 10 repetitions.



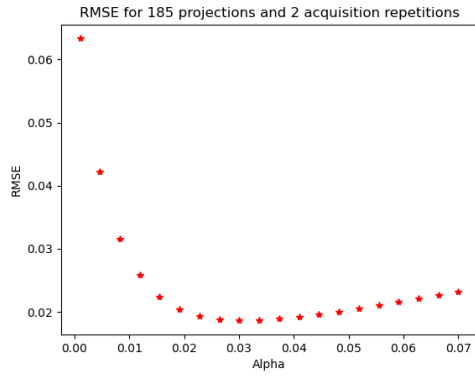
(b) 10 projections and 10 repetitions.



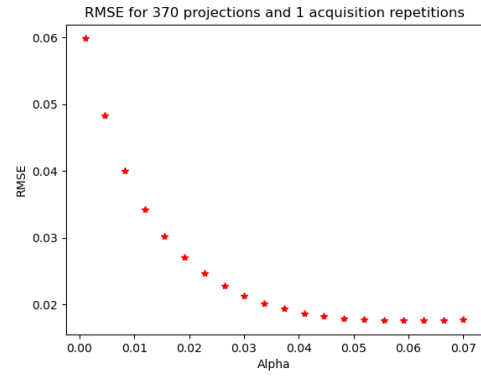
(c) 37 projections and 10 repetitions.



(d) 74 projections and 5 repetitions.



(e) 185 projections and 2 repetitions.



(f) 370 projections and 1 repetitions.

Figure 6.9: RMSE for different α values.

The optimal choice of α does as expected depend on the number of projections, with a greater level of regularisation required for a larger number of projections. The most suitable α values are determined by finding the minimum for each projection. The results

can be seen in table 6.5.

Table 6.5: Results from testing optimal α .

Projections	Optimal α	Ratio
5	0.0192	261.0
10	0.0083	1210.2
37	0.0083	4477.7
74	0.0155	4766.1
185	0.0301	6155.9
370	0.0591	6260.0

The table also show the ratio between the number of projections and the optimal α calculated by:

$$\text{Ration}(\text{Projection}) = \frac{\text{Projection}}{\text{Optimal } \alpha} \quad (6.3)$$

When the number of projections increases the system matrix $\mathbf{A}$ presented in section 3.2 will increase in number of rows since a greater total number of rays are captured.

If the optimal α is directly proportional to the size of $\mathbf{A}$, and sequentially the number of projections should the ratio from table 6.5 be roughly constant. The ration does not appear to be constant and a linear trend is thereby not likely. It is therefore concluded that an full 'sweep' of α , similarly to the one conducted, must be performed for each configuration to identify the optimal α value

6.5 Best Acquisition Parameters

With the conclusion on how to find the best regularisation parameter and a ground truth image, the best reconstruction for each configuration can be evaluated. The photon count is kept constant in conformity with the configurations seen in table 6.4. Again is 10 acquisition repetitions used for both 5, 10, and 37 projections due to the low number of repetitions. A reconstruction is made with the best α and compared to the ground truth reconstruction found in 6.3.2 using the root mean squared error (RMSE) from 5.1. The error is found at channel 30 (70.59 keV) and the results can be seen in figure 6.10.

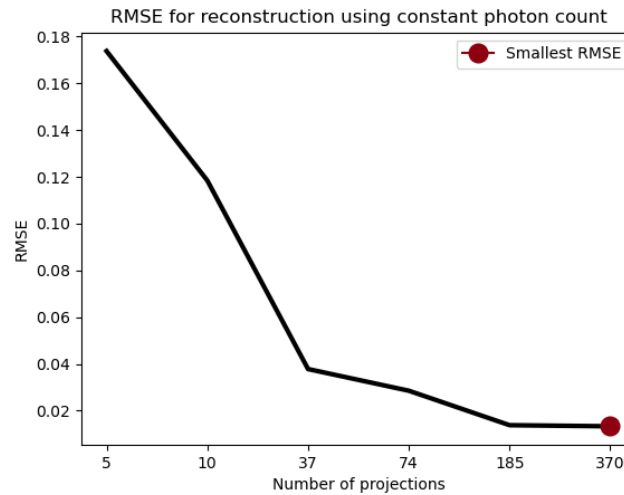
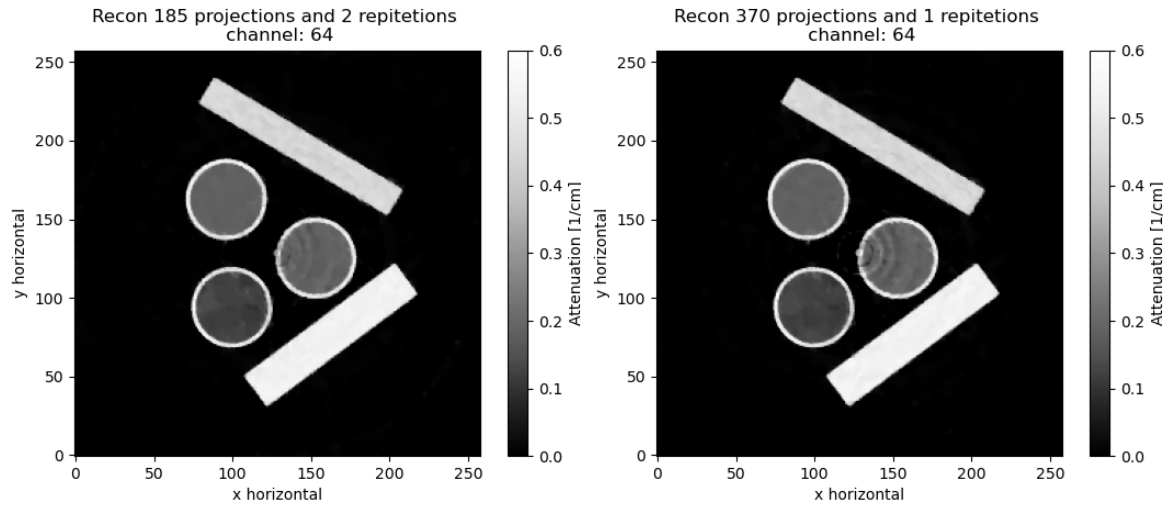


Figure 6.10: RMSE for reconstructions using constant photon count

The RMSE decreases when the number of projection increases even with the signal quality becoming lower due to the decreasingly number of repetitions. The lowest error can be found at 370 projections. It is also notable that the error only decreases slightly between 185 projections and 370 projections even though the number of projections is doubled. It is therefore very likely that the RMSE will have a minimum if the number of projections is increased. This is unfortunately not possible to achieve with the current data set. The reconstruction for 185 and 370 projections can be seen side by side in figure 6.11. It can actually be seen that ring artefacts starts to emerges at 370 projections. This is due to low signal quality for when only 1 repetition is used at 370 projections.



(a) Reconstruction using 185 projections and 2 repetitions. (b) Reconstruction using 370 projections and 1 repetitions.

Figure 6.11: Best reconstruction for respectively 185 and 370 projections. Seen at channel = 64.

Based on the experiments conducted is concluded that the utilised data set does not have enough projections to gain conclusive results regarding the best acquisition parameters. A new data set will therefore have to be designed and captured. With that said have the this preliminary study still been fruitful, and we can use it to conclude that an full α 'sweep' will have to be conducted for each projections/repetitions configuration to decide optimal regularisation parameter.

Before a new data set can be made, will we investigate another reconstruction method.

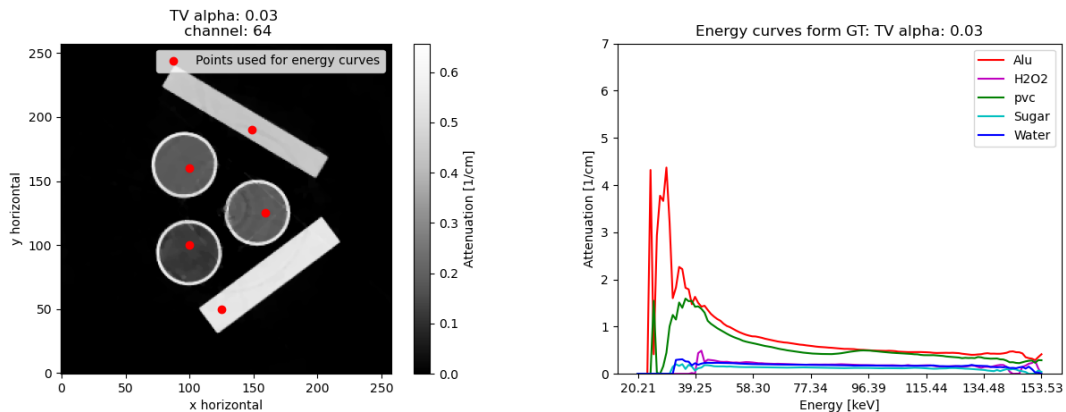
CHAPTER 7

Regularisation Along the Energy Dimension

During the course of this project, I had the opportunity to participate in a CIL (Core Image Library) training session and hackathon event in Cambridge, organised by the Collaborative Computational Project in Tomographic Imaging (ccpi). This event gave me the opportunity to work with my data in ways, which would otherwise had extended beyond the original scope of the project. The data worked with is the existing data set presented in section 6.1.

7.1 Construction of Energy Curve

In our prior investigations, regularisation has solely been employed in the spatial dimension, resulting in distinct solutions for each individual energy-channel. We can get a quick look at the attenuation profiles along the energy dimension by extracting the attenuation values for a specific material within the reconstruction for each channel. For simplicity we will look at the simple approach involving the use of a singular pixel from each material to construct the curves. Firstly we look at the ground truth reconstruction seen in figure 6.8a, and the points have been found and the energy curve for each material can be seen in figure 7.2.



(a) Ground truth reconstruction with points where energy curves are found.

(b) Energy Curves for ground truth.

Figure 7.1: Ground truth reconstruction where single points have been used to find energy curves for each material.

It is seen that the curves are very rough and non-smooth. A portion of this can be due to that fact that only one pixel from each material have been extracted. The curves are also especially uneven in the lower energy channels due to some of the detector distortion described in 4.1. It is also seen that there are no signal in the first few channels, and the attenuation is 0 as a result of this.

7.2 Setting up Regularisation in the Energy Dimension

In order to get smoother energy curves a regularisation term needs to be defined in the energy dimension. The regularisation used in the space dimension and the energy dimension does not have to be the same. The optimisation problem is formalised as:

$$u^* = \arg \min_u \|\mathbf{A}\mathbf{u} - \mathbf{b}\|_2^2 + \alpha \cdot J_{x,y}(\mathbf{u}) + \beta \cdot J_e(\mathbf{u}), \quad (7.1)$$

where $J_{x,y}(u)$ is a regularisation term for the space dimension and $J_e(u)$ is a regularisation term for the energy dimension. α and β are regularisation parameters. Total variation (TV) will be used in the space dimension with same parameters as in the ground truth reconstruction shown in figure 6.8 ($\alpha = 0.03$). In the energy dimension will Tikhonov regularisation as described in section 3.4.1 be employed. The reasoning is that Tikhonov is good for smoothing compared to TV which is good for maintaining sharp edges. The optimisation problem then becomes:

$$\min_u \|Au - b\|_2^2 + \alpha \cdot \text{TV}(u) + \beta \cdot \|D_e \mathbf{u}\|_2^2 \quad (7.2)$$

The problem will be solved using the Primal Dual Hybrid Gradient method (PDHG), which is a powerful splitting method used for optimisation problems with multiple terms. The method has been implemented in CIL described [21], and the implementation of the problem in eq: (7.2) has been done following the approach in [21].

7.3 Different Choices of β

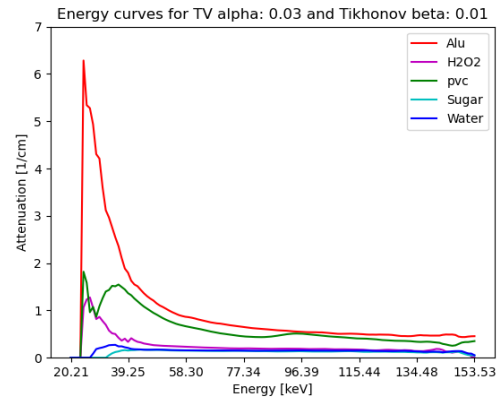
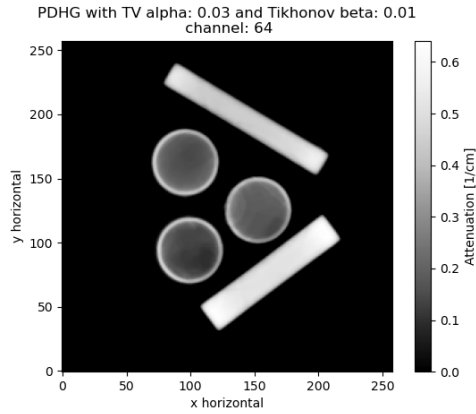
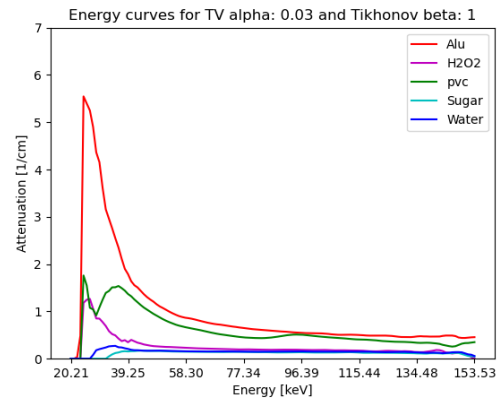
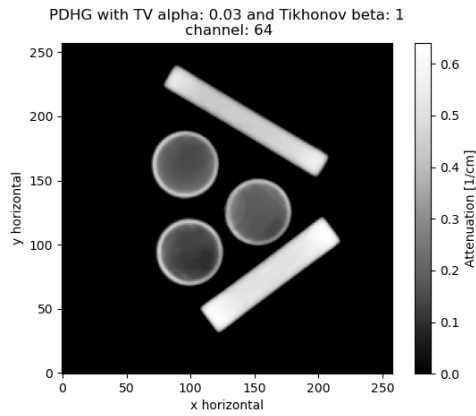
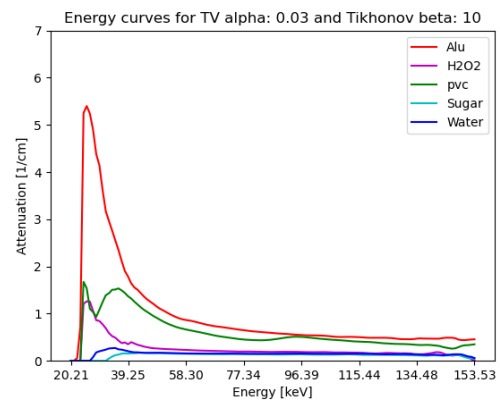
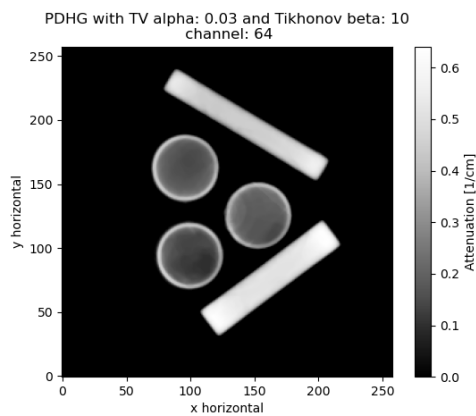
Just as with the TV-regularisation will the choice of regularisation parameter yield different results. The parameter associated with TV α will be kept constant at 0.03, since this has already been found to be optimal for the data. Different choices for the parameter associated with Tikhonov, β will be tested:

$$\beta = [0.01, 1, 10] \quad (7.3)$$

The energy curves are then found using the same points as illustrated in 7.1a, and the resulting energy curves together with the reconstructions can be seen in figure 7.2.

It can be seen that the curves appears more smooth as β increases. The large inconsistencies at lower energy channels also seems to be removed. The attenuation still appears as zero in the first energy channels. It is also notable that the aluminium curve becomes lower with larger regularisation parameter β . This is also to be expected since regularisation generally evens out the solution.

Only a few chooses for β in a relatively large span have been tested, and more analysis will of course need to be performed to find an optimal β . But it can still be concluded that regularisation in the energy dimension will have an impact on the attenuation curves that can be drawn from the reconstruction. A Jupyter notebook detailing this small side project have been created with the purpose to be published for interested CIL users. The notebook will be published in the CIL user showcase: <https://github.com/TomographicImaging/CIL-User-Showcase>

(a) Reconstruction using α : 0.03 and β : 0.01.(b) Energy Curves using α : 0.03 and β : 0.01.(c) Reconstruction using α : 0.03 and β : 1.(d) Energy Curves using α : 0.03 and β : 1.(e) Reconstruction using α : 0.03 and β : 10.(f) Energy Curves using α : 0.03 and β : 10.**Figure 7.2:** PDHG solution using α : 0.03 for TV and various β for Tikhonov.

CHAPTER 8

Data set with more projections

The current data set utilised until now only consist of 370 projections. As concluded in section 6.5 is this not enough, since the limited number of projections constrains the options for investigating low-numbered projections, as a choice has been made to exclusively employ factors of 370 during downsampling, as depicted in (6.1). The choice to create a new data set have therefore been made.

8.1 Design of the Data Set

The design of the data set aims to incorporate a highly factorisable number of projections. In the study conducted in [22], it was determined that the highly composite number 2520 serves as an optimal choice for the number of projections. This selection results in the generation of a large number of perfectly equiangular sub-sampled data sets. Notably, the number 2520 possesses many factors within the range of 20 to 10. This is desirable considering that the objective of this research is to emulate baggage scanning data, which mostly operates within this range. The dimension of the data acquired can then be seen in table 8.1. It can also be seen from the table that 1 vertical slice and 128 energy channels are still used. The geometry employed are still a 2D fan beam, which is illustrated in figure 8.1.

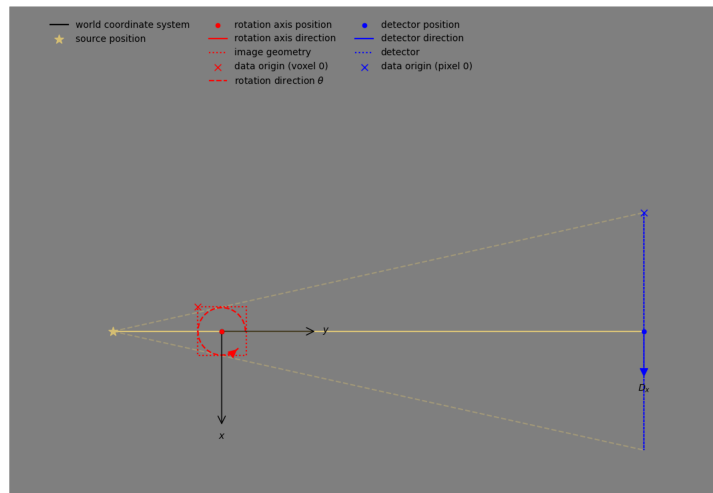


Figure 8.1: Geometry set-up for new data set.

Table 8.1: Dimension of data used.

Projections	Vertical slice	Horizontal slice	Energy channels
2520	1	640	128

Table 8.2: Acquisition parameters.

Projections	Angular span	Source Detector distance	Source - Axis Of Rotation distance
32520	360 [Degrees]	114.8 [cm]	23.5 [cm]

The acquisition parameters can be seen in table 8.2.

With the number of projections determined, the number of acquisition repetitions also needs to be determined. If we want to follow to same factorisation used in section 6.4.1 the following repetitions that should be used for each projection can be seen in tabel 8.3.

Table 8.3: Configurations of projections and repetitions.

Projections	2520	1260	840	630	504	420	360	315	280	252
Repetitions	1	2	3	4	5	6	7	8	9	10
Projections	210	180	168	140	126	120	105	90	84	72
Repetitions	12	14	15	18	20	21	24	28	30	35
Projections	70	63	60	56	45	42	40	36	35	30
Repetitions	36	40	42	45	56	60	63	70	72	84
Projections	28	24	21	20	18	15	14	12	10	9
Repetitions	90	105	120	126	140	168	180	210	252	280
Projections	8	7	6	5	4	3	2	1		
Repetitions	315	360	420	504	630	840	1260	2520		

Projections under 9 is not useful to examine, so 280 repetitions should be enough. Therefore will the data set include 40 configurations corresponding to the first 4 rows in table 8.3. Each repetition is only at 2 ms to create a noisier data set compared to the one previously used.

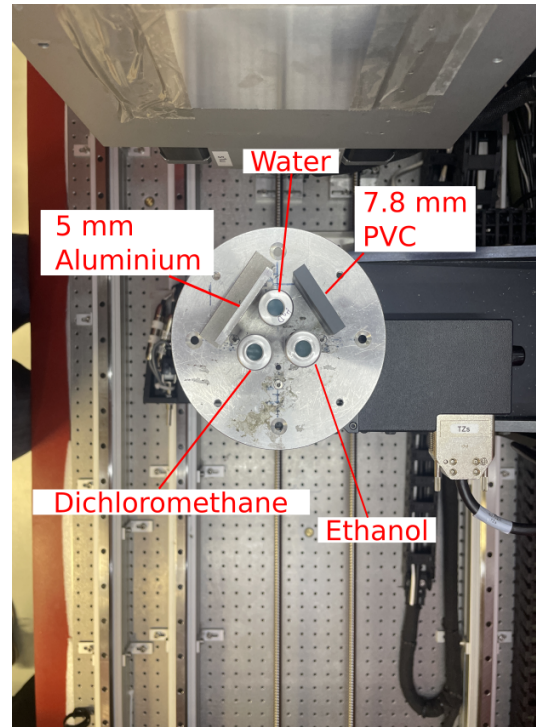
8.1.1 Experiment Equipment

The detector used is still the Multix ME100 photon counting detector with 128 energy channels [20]. Now the experimental set-up consist of 5 detector modules each of 128 pixels chained together with a small gap between. The gap is measured to be roughly 3 pixel wide pr. gap. A picture of the detector is seen in figure 8.2a. The 5 modules

and the 4 gaps between the modules can easily be seen. The X-ray source is still the Hamamatsu Microfocus X-ray source L12161-07 with a tungsten anode.



(a) Detector used with 5 modules chained together.



(b) The 5 materials.

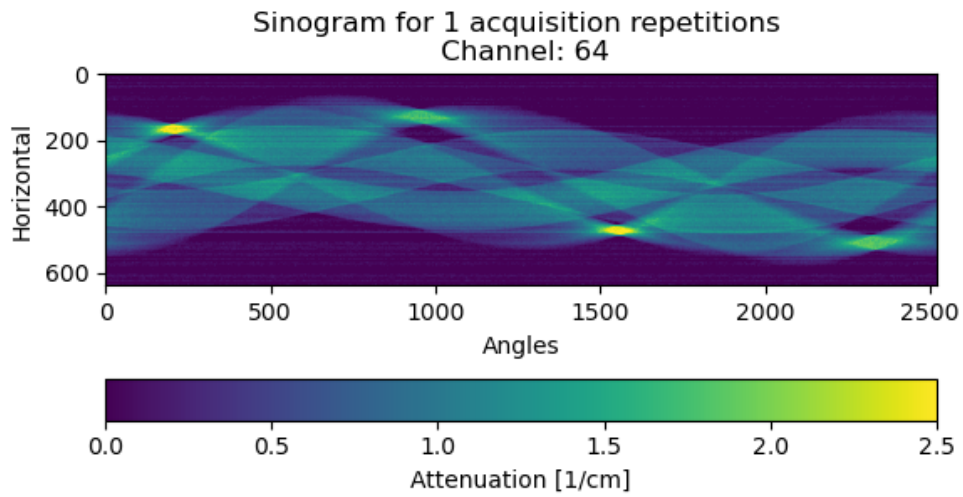
Figure 8.2: Detector and material setup.

8.1.2 Materials

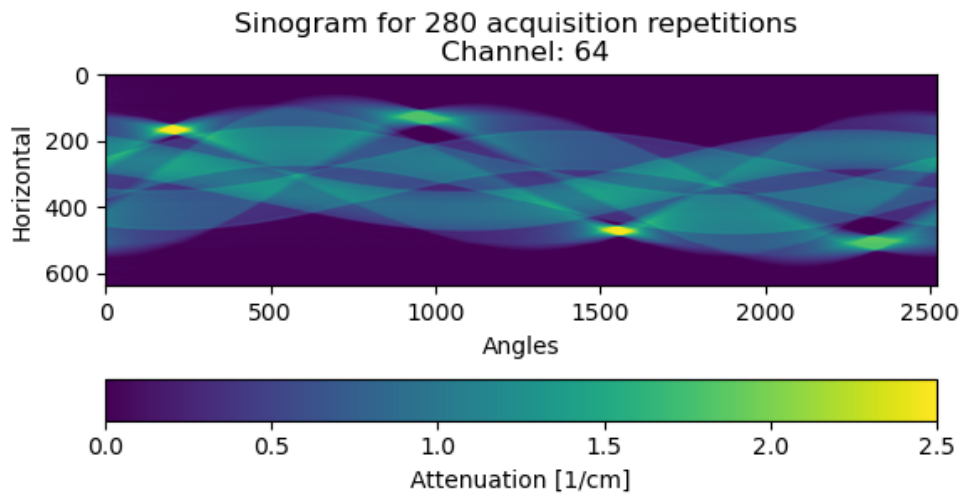
Just like the preliminary data set the object of interest consist of 5 materials, which is; Ethanol (C_2H_6O), Dichloromethane (CH_2Cl_2), Water (H_2O), 5 mm aluminium, and 7.8 mm PVC. The materials Ethanol, Dichloromethane and Water are placed in small glass containers. The placement of the materials can be seen in figure 8.2b.

8.1.3 Preprocessing of Sinograms

40 different sinograms have been corrected and made using the correction algorithm described in 4.1. Two of the the sinograms can be seen in figure 8.3 for 1 repetition and 280 repetitions respectively. It is clearly seen that a higher number of repetitions creates a less noisy signal. It is also seen that no dead/hot pixels seems to appear in the sinograms, and a processing step to remove these is thereby not needed.



(a) Sinogram for 1 repetition.



(b) Sinogram for 280 repetition.

Figure 8.3: Sinograms for new data set.

The detector gaps describe in section 8.1.1 will be handled using the same method already described in section 6.2.2, where blank pixels corresponding to the gap width will be inserted and interpolated using linear interpolation. The only difference is that the data now contains more gaps that will need to be interpolated.

8.2 Ground Truth Reconstruction

In continuation, a separate data set has been developed specifically for the purpose of generating an accurate reconstruction to use as a ground truth. The experimental setup employed for this data set is identical to the original, utilising a total of 2520 projections. To minimise the impact of noise 50 repetitions have been used, with each repetition lasting 100 ms per projection.

8.2.1 Iterations study

Until now have an exceedingly large number of iterations been used when creating the ground truth to be certain of the convergence in the solution. However, considering the size of the data set and the interest of not having a time consuming reconstruction an investigation on how many iterations are sufficient is performed. Convergence of a solution is achieved when the objective value of the optimisation problem reaches a state of stability, since the algorithm can not find better a better solution. Our analysis will therefore focus on assessing the objective values from multiple reconstructions using a different number of projections. The reconstruction have been found using TV-FISTA with a regularisation parameter of alpha 0.03. Figure 8.4 shows the objective values for the tested projections which are:

$$9, 35, 70, 210, 2520 \quad (8.1)$$

The objective values have been divided with the number of projections to make comparing easier, since more projections naturally leads to a larger objective value due to the scaling of the system matrix $\mathbf{A}$.

The figure shows that the solution converges quickly. It is also seen that fewer projections leads to more iterations before the objective function is stable. This is to be expected. Therefore should the number of iterations that is sufficient for 9 projections also be sufficient for a higher number of projections. The objective values looks to stabilise between 100 and 200 iterations. Figure 8.5 displays the reconstruction after respectively 10, 100, and 200 iterations for 9, 70, and 2520 projections. It can be seen that the reconstruction only changes slightly between 100 and 200 iterations. We will therefore use 200 iterations for making the ground truth and also in other reconstructions in the following experiments.

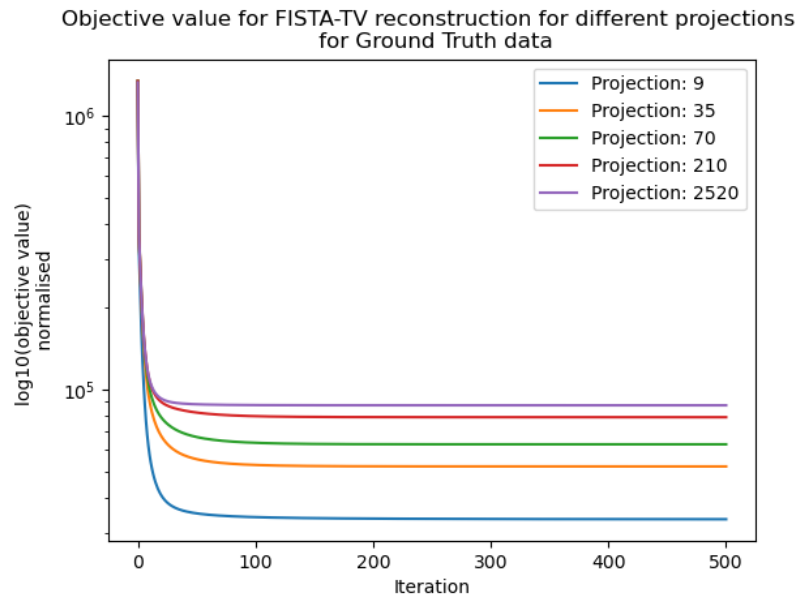


Figure 8.4: The objective values when reconstruction with different projections.

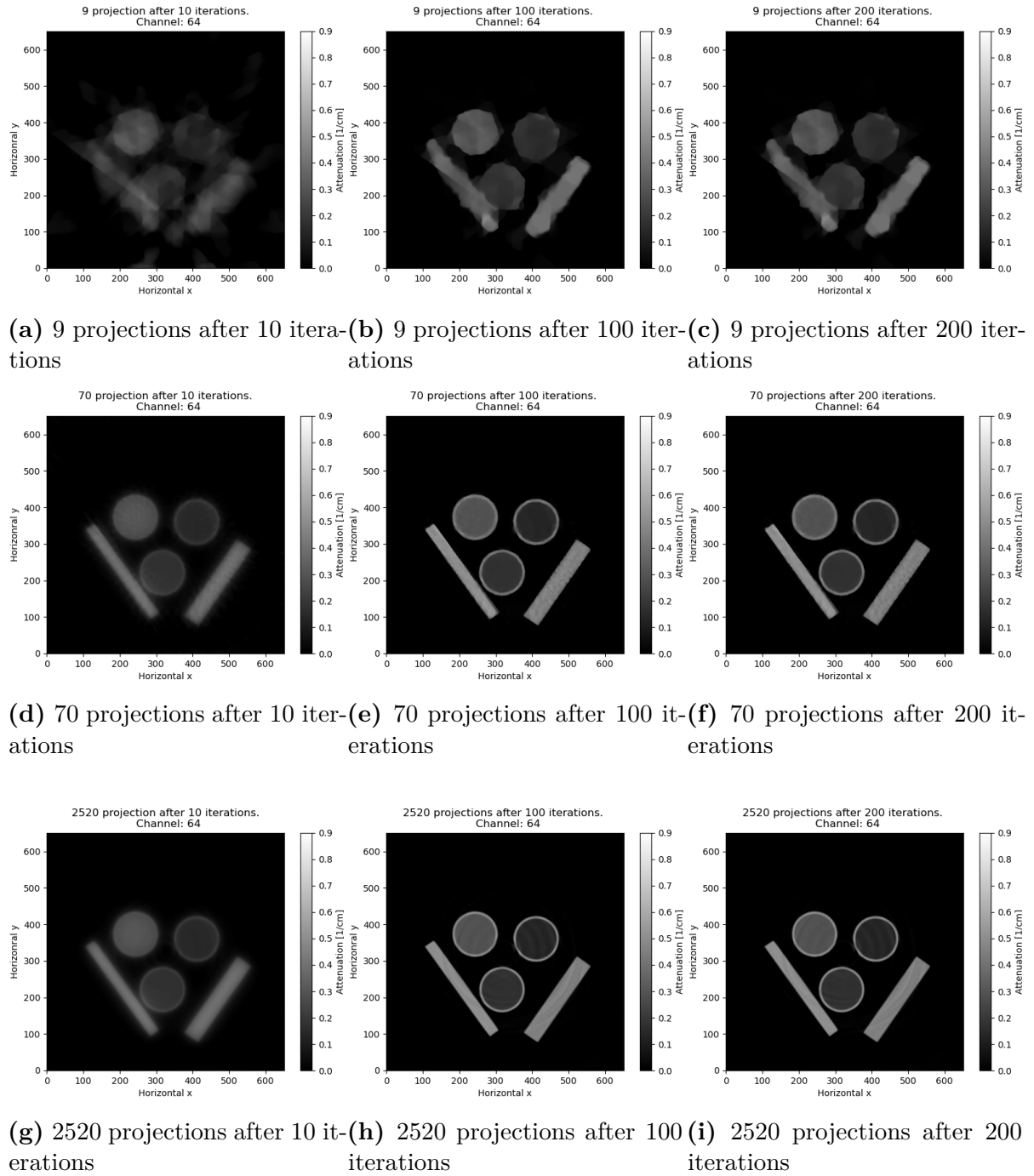
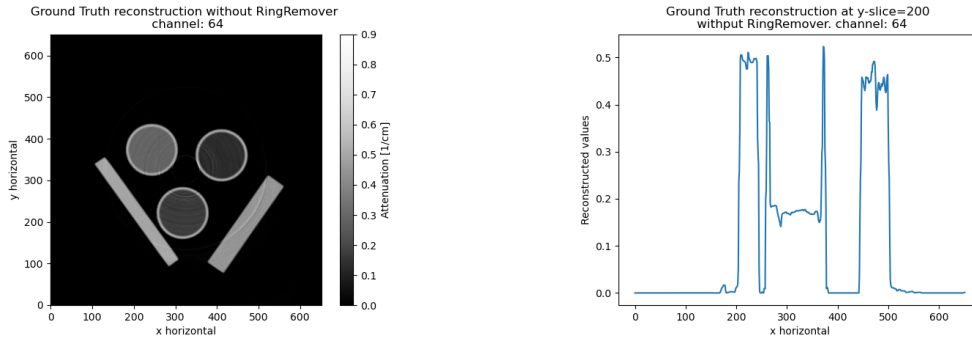


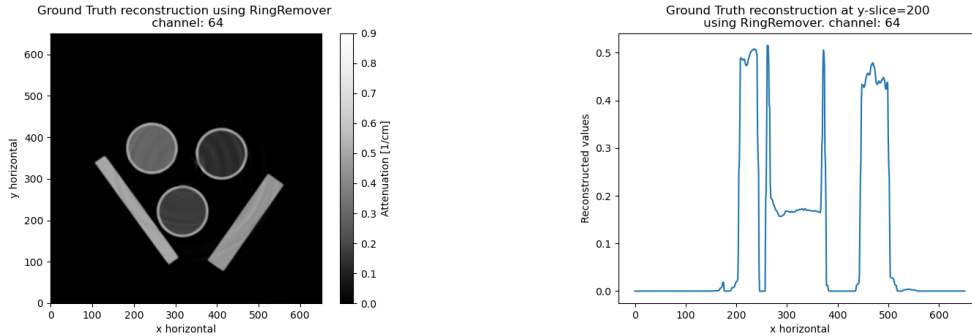
Figure 8.5: Reconstruction of ground truth data set using FISTA-TV after various iterations.

The final ground truth reconstruction can then be generated using 200 iterations and $\alpha = 0.03$. Different α values have been tested and $\alpha = 0.03$ have been chosen based on visual examination of the reconstruction and a profile plot at $y = 200$ similar to the study conducted in section 6.3.2. The resulting ground truth reconstruction at its profile plot is seen in 8.6. It can be seen that the reconstruction have some unwanted ring artefacts. These can also be seen in the profile plot. We will therefore use the **RingRemover** function from CIL described in section 4.4 to smooth out the rings.



(a) Ground truth reconstruction with $\alpha = 0.03$. (b) Profile plot for the reconstructed image at $y=200$.

Figure 8.6: Ground truth Ground truth reconstruction with $\alpha = 0.03$.



(a) Ground truth reconstruction with $\alpha = 0.03$. (b) Profile plot for the reconstructed image at $y=200$.

Figure 8.7: Ground truth Ground truth reconstruction with $\alpha = 0.03$ and **RingRemover** used.

Figure 8.7 shows the ground truth reconstruction with same parameter $\alpha = 0.03$ after the use of **RingRemover**. The rings are not completely erased in the resulting reconstruction, but they are more smooth. This can also be seen from the profile plot. The ground truth reconstruction used for the remaining experiments is therefore chosen to be the reconstruction presented in figure 8.7 where **RingRemover** have been utilised.

8.2.2 Ground Truth Energy Curves

We will also create a set of ground truth energy curves to be able to use the figure of merit based on energy curves presented in section 5.2. We will use the ground truth reconstruction seen in figure 8.7 to construct the energy curves from. We will use the two pixel sampling methods presented in 5.2.2, *Random Pixels* and *Small Area*. The pixels sampled in the ground truth reconstruction can be seen in figure 8.10 for the 2 materials.

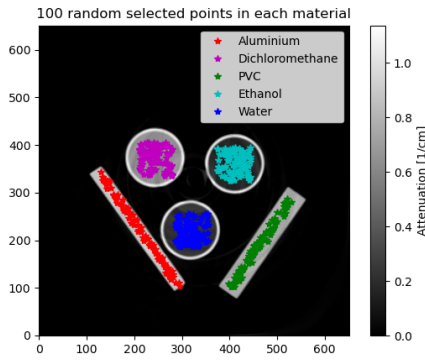


Figure 8.8: 100 pixels random selected in each material

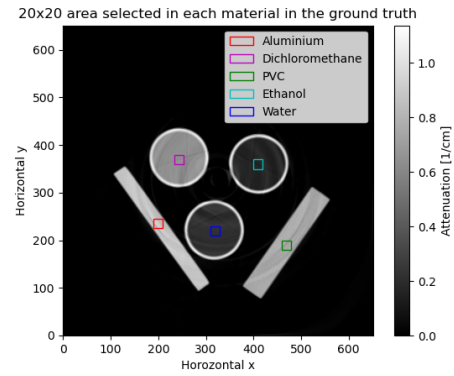


Figure 8.9: 20x20 area selected in each material

Figure 8.10: sampled pixels in each material

It can be seen that the method *Small Area* is restricted by the thickness of the aluminium plate, since we want to sample the same amount of pixels from all the materials. The method *Random pixels* captures pixels from more varying parts of the materials, even though it uses less pixels. The energy curves can then be found by averaging the sampled pixels for each material. The curves in the energy span $[40.15 - 103.14]$ keV (channel: 20-80) can then be seen in figure 8.13.

The two curves look roughly the same with only small deviations. The curves can be compared to the table value curves in the same energy span from figure 5.1b. It is seen that grouping of ethanol and water being in the bottom and aluminium, PVC, Dichloromethane being in the top is correct. Dichloromethane does have a higher attenuation in the reconstructed curves compared to the NIST curves. It is also notable that in the ground true curves does PVC have a bump around 90 keV, which is not present in the NIST curves. This is likely due to the presence of a lead stabiliser in the PVC [23], since lead has a k-edge at 88 keV. This is one of the main reasons for using ground truth curves made from data instead of table values curves for the figure of merit. The curves from method *Random pixels* have been chosen as the ground true curves, which will be used for the figure of merit, due to the method's ability to capture the most parts of the materials.

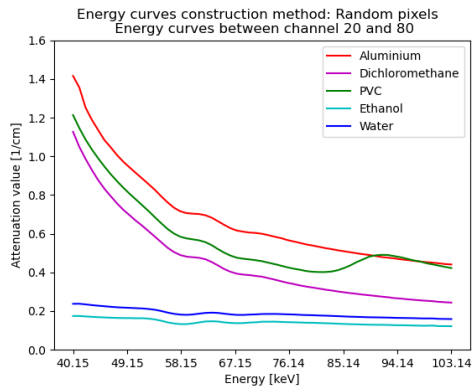


Figure 8.11: Method: *Random pixels*

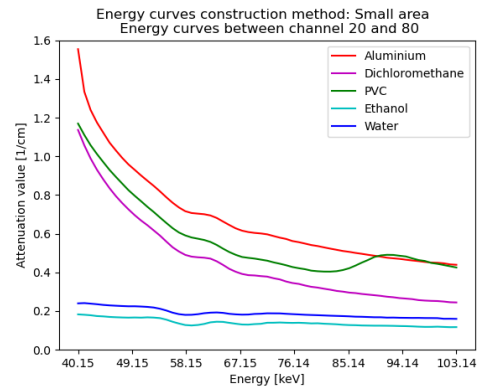


Figure 8.12: Method: *Small Area*

Figure 8.13: Ground true energy curves in the span 40.15 - 103.14 [keV] for the two method

CHAPTER 9

Experimental Study and Results

9.1 Outline of Desired Study

The objective is to conduct an experiment in a manner akin to the study performed in Section 6.5, where we find the constant photon configuration that yield the lowest RMSE when comparing it to the ground truth. Additionally, we want to broaden the investigation by incorporating various figures of merit based on the discovered energy curves, as presented in Section 5.2. One of the primary aims of this study is to examine whether there exists a configuration in which the reduction in signal noise resulting from a smaller number of repetitions outweighs the requirement for a greater number of projections to achieve improved reconstruction quality, when the photon count is kept constant. Furthermore a goal of the study is to explore different figure of merits ability to effectively assess the quality of reconstructions.

9.2 Regularisation Parameter Sweep

Firstly an α sweep similarly to the one made in 6.4.1, where the photon count is kept constant, will need to be conducted to find the best α values to reconstruct with later. The span which α values have been tested in are:

$$\alpha = [0.001, 1.313] \tag{9.1}$$

A distinct α have been found for each configuration, by finding the α that results in the smallest error when using the RMSE measurement described in 5.1 against the ground true image. Plots of the error as a function of α for selected combinations can be seen in figure 9.1.

It can be seen that the optimal α increases with the number of projections. An unambiguous optimal α have been found for each configuration and it is therefore concluded the sweep have been successful.

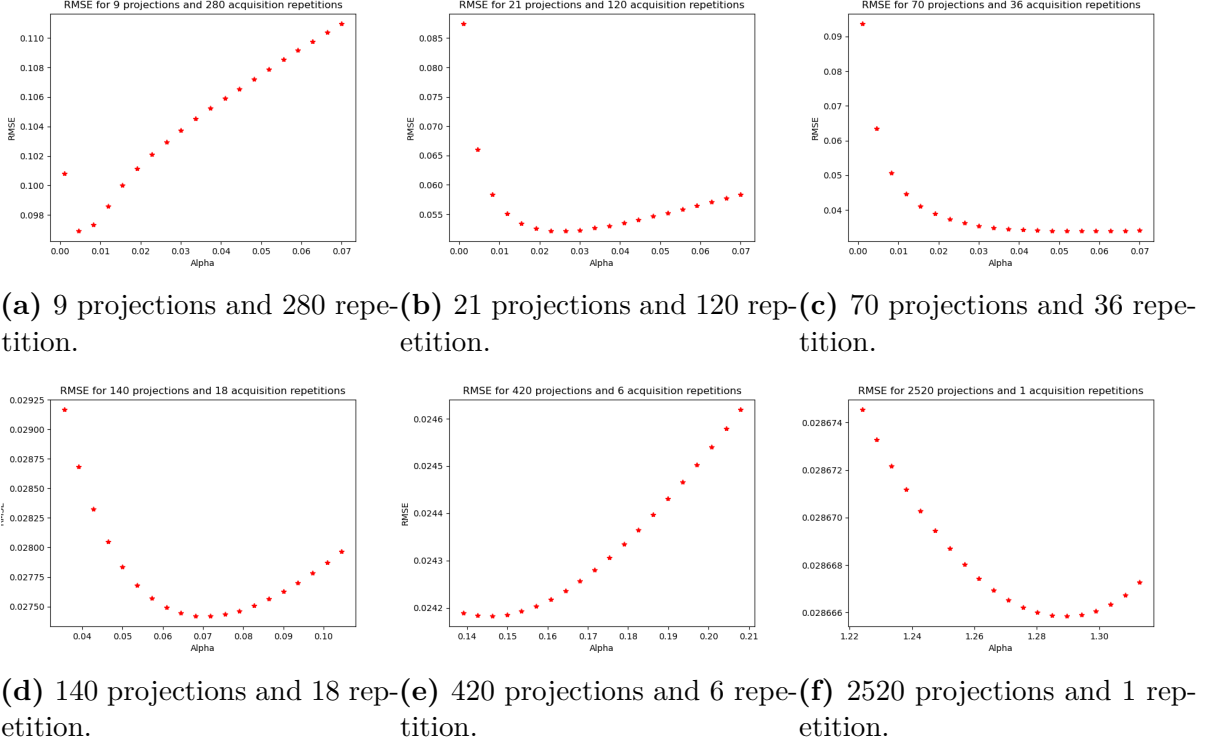


Figure 9.1: Alpha sweep for selected projection/repetitions combinations. The RMSE have been plotted different α -span to emphasise that the minimums are in different regions of the α -span tested.

9.3 Acquisition Experiment Using RMSE as FOM

Each projection/repetitions configuration are reconstructed with the best α . The error for the reconstruction is found the RMSE form eq: 5.1 against the ground truth reconstruction seen in figure 8.7 using channel 30 (70.59 keV). The photon count is kept constant and the resulting error can be seen in 9.2.

It is seen from the plot that the quality of the reconstruction increases with the number of projections. This is to be expected. But it is also seen that the RMSE has a minimum, which is found to be 0.0242 at 420 projections and 6 acquisitions repetitions. This means that there is a point where more acquisitions repetitions is more important than more projections, and the ration between them is optimal at 420 projections. Figure 9.3 shows the reconstructed image for selected configurations. It can be seen that visually the reconstruction using 420 projections is also the better reconstruction. At 2520 projections have the X-ray source only been open 2 ms pr. angle since we only use 1 repetitions, and this equals a lot of signal noise which can also be seen in the reconstruction, where big ring artefacts form.

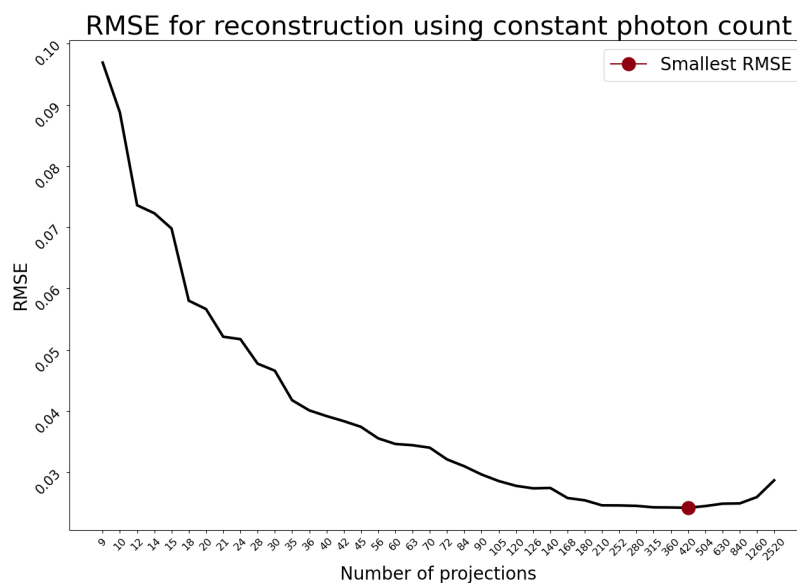


Figure 9.2: RMSE for different projections.

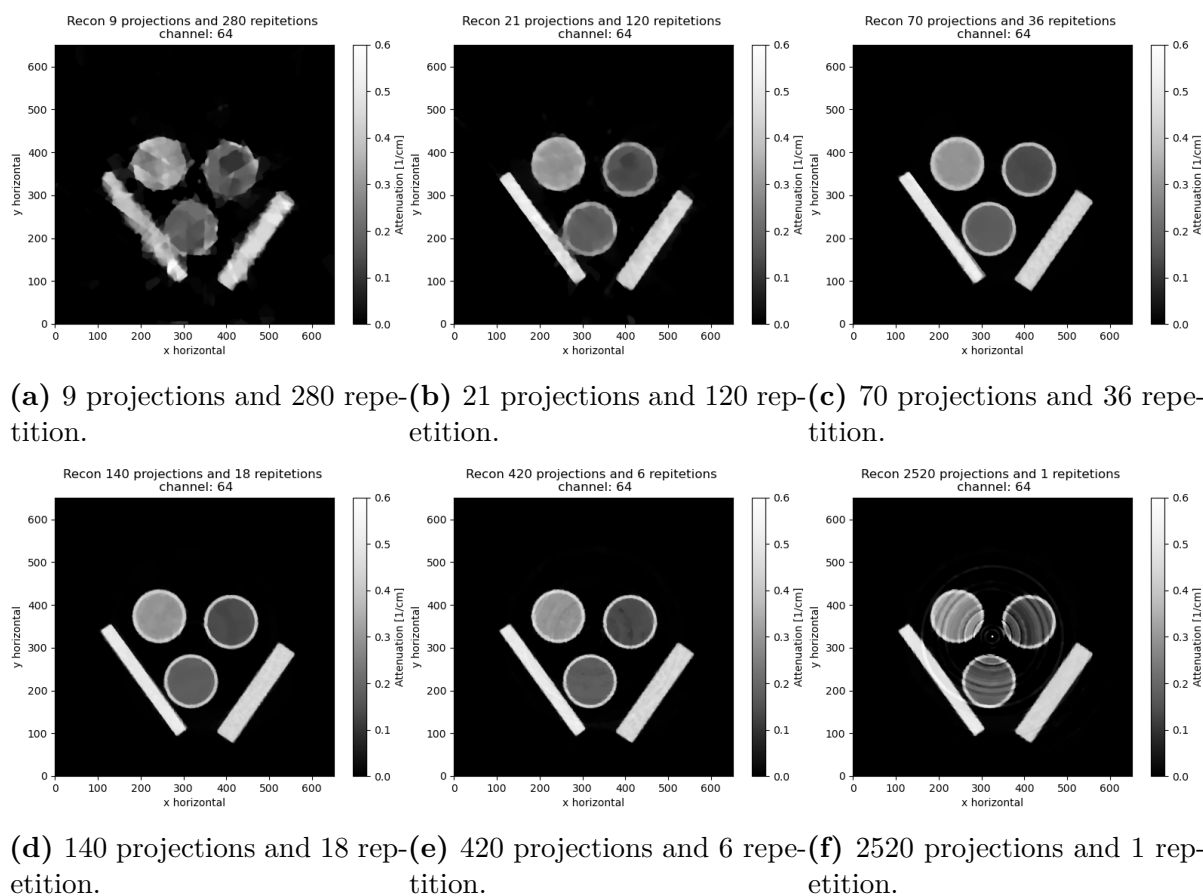


Figure 9.3: Reconstruction for selected projection/repetitions configurations. Shown at channel=64.

9.4 Acquisition Experiment Using the Energy Curves as FOM

The energy curve based figure of merit presented in section 5.2 will also be used to study the error with different acquisition parameters. Specifically will the 4 different error combinations described in table 5.1 be implemented. The ground truth curves used for finding the error are the ones presented in figure 8.11. We can construct the energy curves for every reconstruction by using the two sampling method described in section 5.2.2; *Random pixels* and *Small Area*, and average the pixels to acquire 1 attenuation value pr. channel. The resulting energy curves for selected projections are plotting against the ground truth curves for each material in figure 9.4.

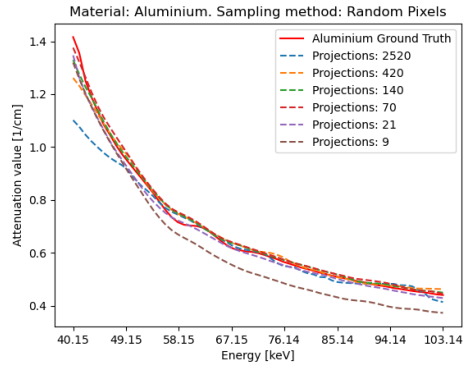
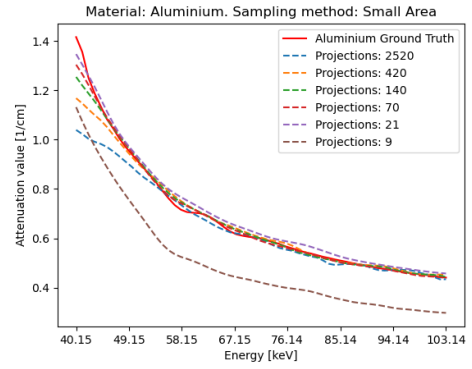
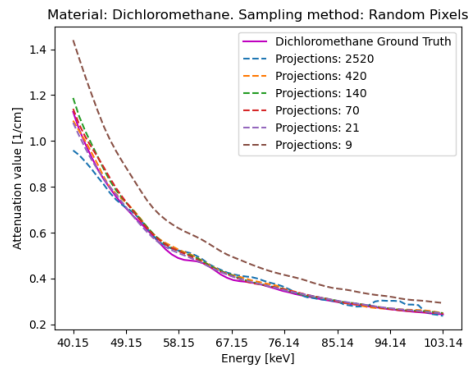
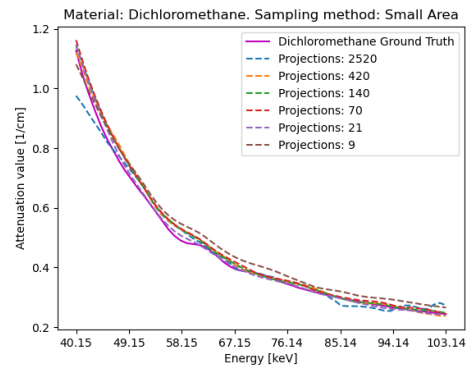
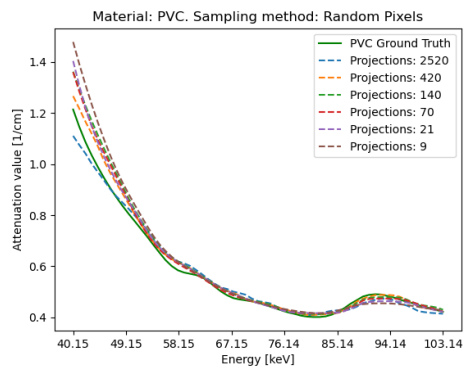
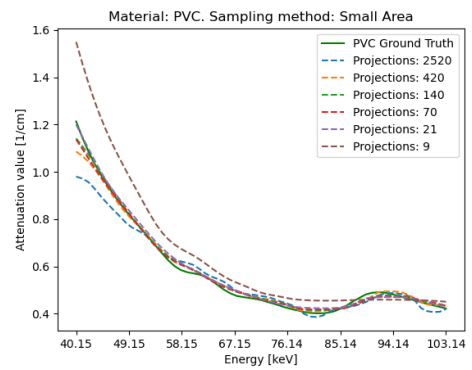
It can be seen that for PVC and Dichloromethane the derived curves closely align with the ground truth curves. Notably does these materials also exhibit different ground truth curves compared to the NIST curves. It is seen that the curves obtained from 9 projections often lie below the ground truth values, as evidenced in both the cases of aluminium and the *Small Area* sampling for Ethanol and Water. Additionally do the curves derived from 2520 projections exhibit significant volatility when compared to the ground truth curves for ethanol and water. It is important to note that to construct the curves the sampled pixels have been averaged for each channel before plotting. In some of the tested error measures will the pixels not be averaged before finding the errors.

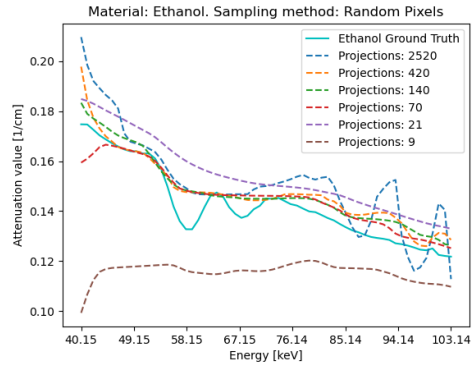
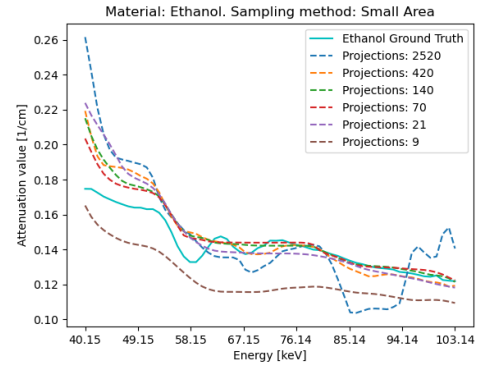
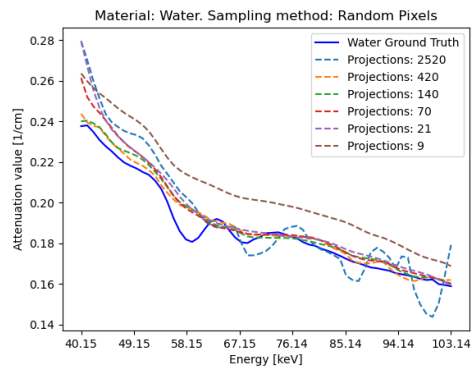
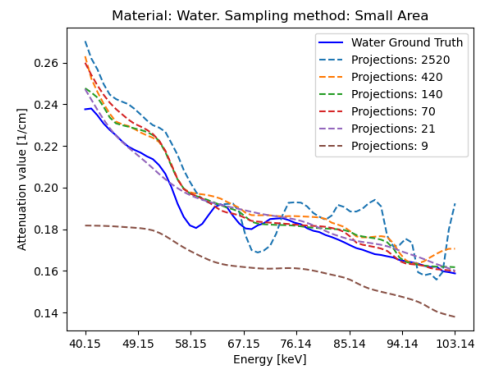
The error at each energy channel will then be found using the 4 different measures presented in section 5.2 and table 5.1. In order to compare the error for different configuration is it desirable to calculate the error as a single number pr. material instead of a whole curve. The channel-wise errors will therefore be summed in the channels: 20 – 80:

$$Error = \sum_{i=20}^{80} = Error(channel = i) \quad (9.2)$$

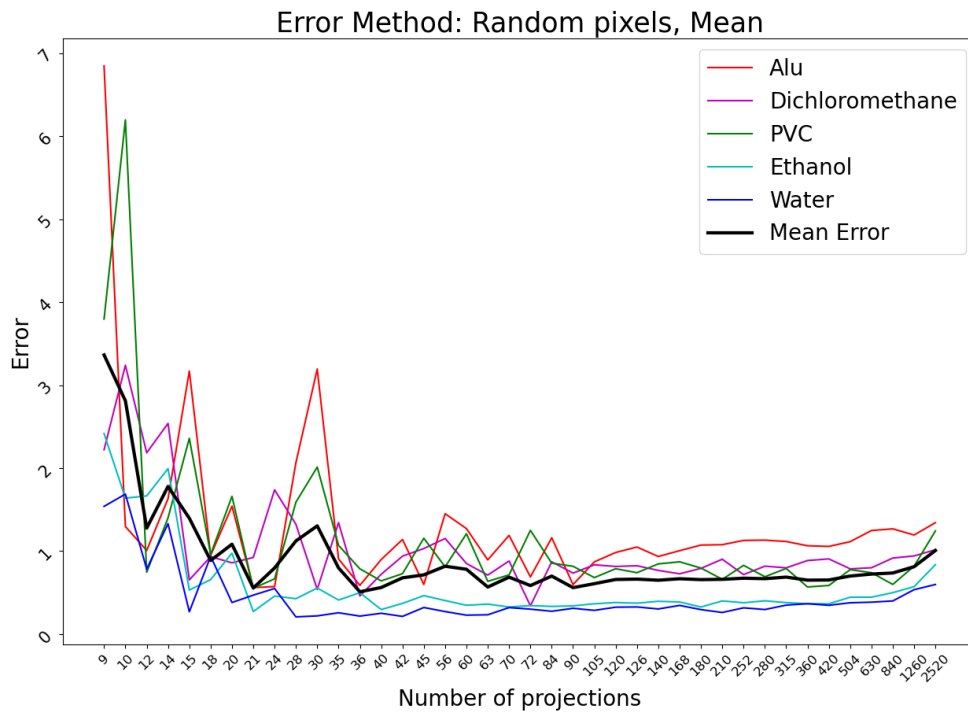
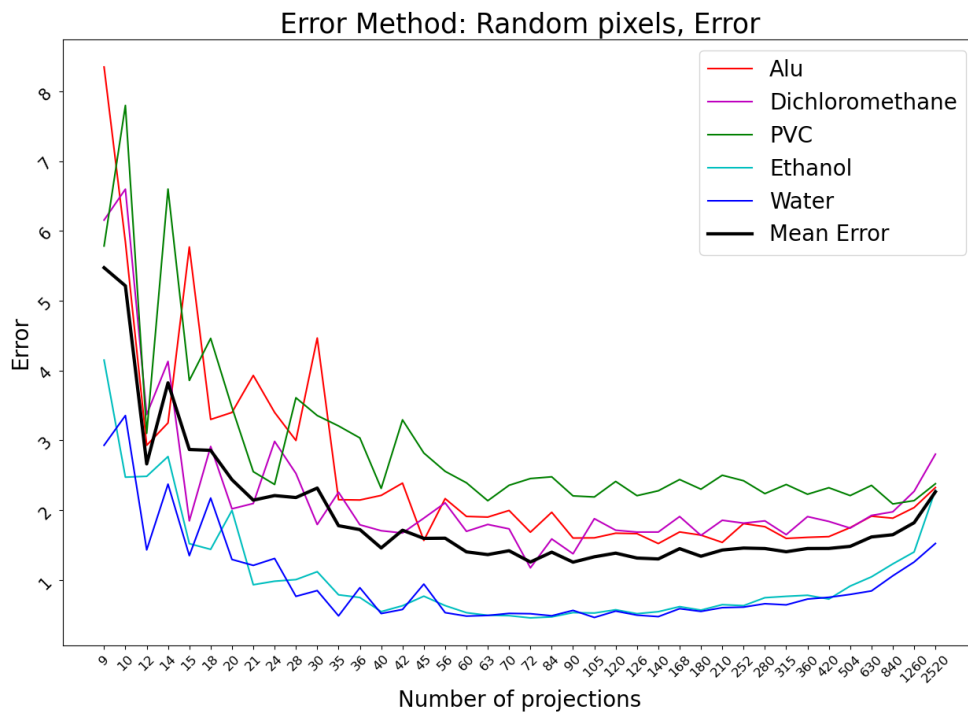
This can be done for each of the 5 materials in the sample. The resulting errors for each projection/repetitions configuration for each of the 4 methods is then presented in figure 9.5. The mean error for the materials is also shown and is used to determine the generally lowest error. The configuration that each error method finds as the best can be seen in table 9.1 together with the error. The observed outcomes reveal that the four methods yield distinct minima. The methods finding the error before averaging obviously have larger error values.

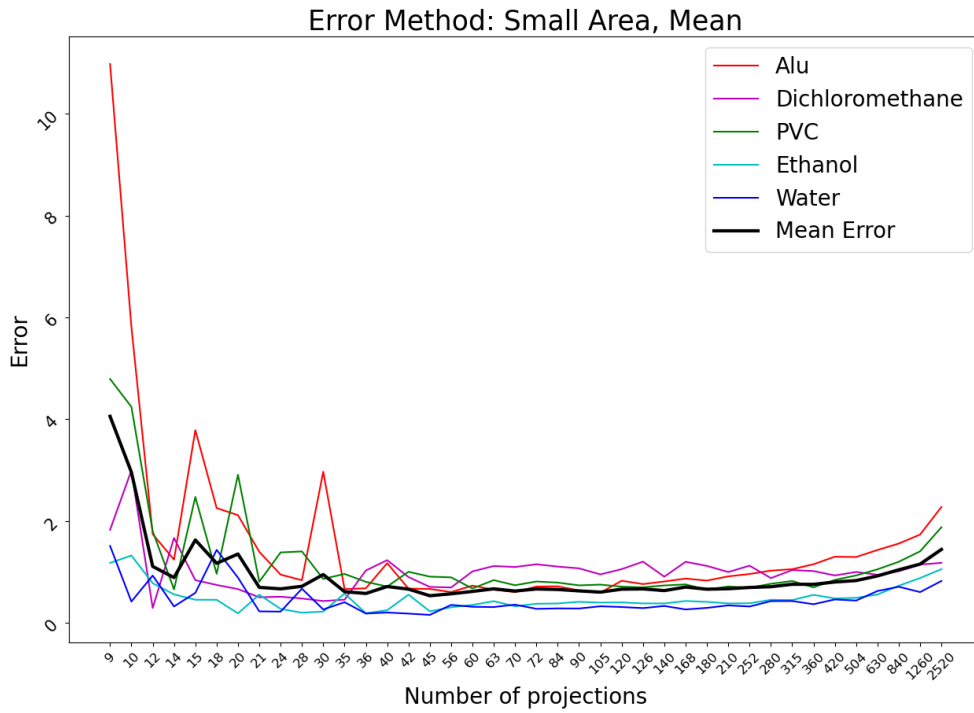
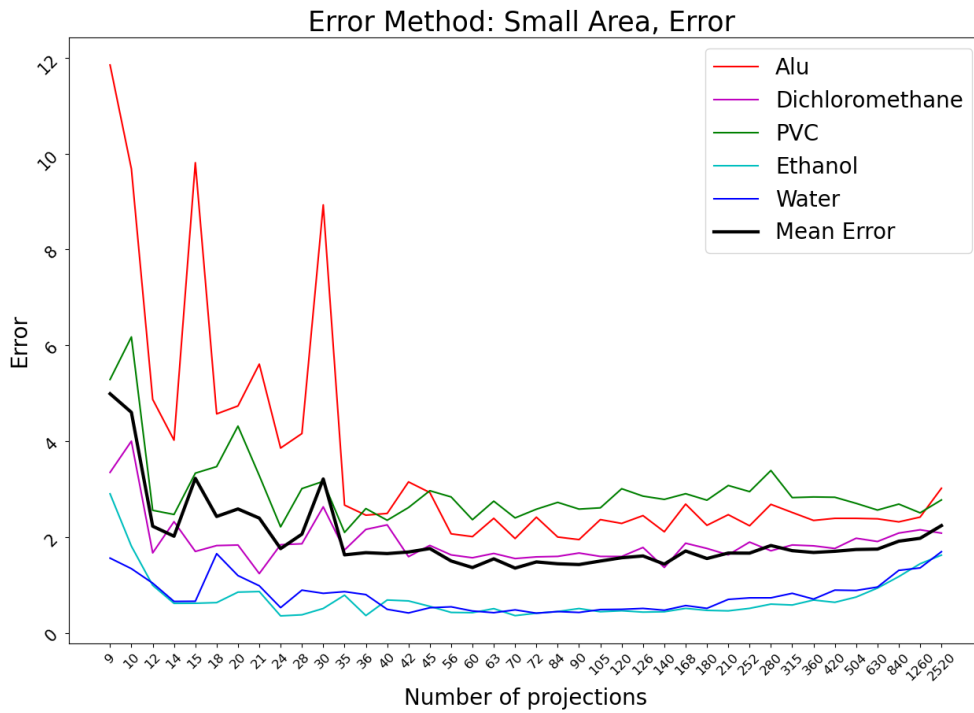
Upon examining the shape of the plots seen in Figure 9.5, is the identification of a minima is not not clearly, instead does it appears that the error tends to flatten after reaching a certain number of projections. Moreover, is a marginal increment in the error seen as the number of projections increases. Notably is it that the plot corresponding to the *Random Pixels, Error* method exhibits the most distinct and well-defined minimum.

(a) Aluminium: *Random Pixels*.(b) Aluminium: *Small Area*.(c) Dichloromethane: *Random Pixels*.(d) Dichloromethane: *Small Area*.(e) PVC: *Random Pixels*.(f) PVC: *Small Area*.

(g) Ethanol: *Random Pixels*.(h) Ethanol: *Small Area*.(i) Water: *Random Pixels*.(j) Water: *Small Area*.**Figure 9.4:** Energy curves for different projections for the two 2 sampling methods.**Table 9.1:** The projection with smallest error for the 4 different methods

Sampling \ Method	Mean $\rightarrow$ Error	Error $\rightarrow$ Mean
Random Pixels	36 Error: 0.51	90 Error: 1.26
Small Area	45 Error: 0.54	70 Error: 1.35

(a) *Random Pixels, Mean*(b) *Random Pixels, Error*

(c) *Small Area, Mean*(d) *Small Area, Error***Figure 9.5:** The error for different projections for the 4 error methods

CHAPTER 10

Discussion

10.1 Discussion of Results

10.1.1 The Figure of Merit

Various figure of merits have been employed within this report to assess the quality of reconstructions with vastly different results. The RMSE measure finds 420 to be the optimal projections when the photon count is kept constant. This FOM is heavily focused on the shape of the elements in the image, facilitated by a direct comparison between the reconstruction and a ground truth image. The ground truth image, utilising a large number of projections, allows for an assumption that the shapes of the elements within the reconstruction closely align with reality. A slight change in the shape of the reconstruction will therefore be penalised a lot in the RMSE. This could maybe be counteracted by implementing a weighting in the RMSE where areas of interest can affect the error more, while area such as the background will be less present in the error calculations. However, it is noteworthy that this error metric has succeeded in identifying a minimum at which increasing the number of projections is outweighed by the loss in signal noise, which is also one of the main goals of this project.

The figure of merit based on the analysis of error curves, reveals distinct minima across a range of projections, spanning from projection 36 to 90. Unlike the RMSE metric this figure of merit prioritises the attenuation values over the shapes in the reconstruction. This could explain why a lot of projections is not necessarily required to achieve a low error, and why the measured errors consistently stays low. Only at a very low number of acquisition repetitions, where the signal exhibits substantial noise, do the artefacts introduced during reconstruction significantly impact the accuracy of attenuation values, due to large inconsistencies in the sampled pixels.

The method was still able to achieve in identifying a minimum, aligning with one of the intended goals. Given that the data utilised in this project aims to emulate airport baggage scanning, where the objective involves material classification, the inclusion of a figure of merit based on attenuation curves is highly valuable. This is due to the fact that the attenuation coefficients derived from the reconstructed image are used as crucial parameters in the classification process of materials in baggage. A figure of merit based on attenuation curves thereby enables an effective assessment of the reconstruction's ability to be used for material classification.

However, there still remains opportunities for further refinement and enhancement of this approach. One potential unexplored improvement could involve incorporating the weighting of various energy channels, using prior knowledge that designates certain channels as particularly significant. For instance does the used PVC have a distinct k-edge within the energy span, and the error measure could be designed to prioritise this characteristics more.

10.1.2 Effect of Average Before Error

One of the main differences (aside from pixels sampling) within the energy curve-based error methods, lies in the point at which the sampled attenuation values are averaged: either before or after the error is determined. The act of averaging these values prior to their comparison with the ground truth has the potential to cancel out significant errors that exist both above and below the ground truth curve. Consequently, such an approach may inadvertently eliminate a portion of the actual error that we aim to measure and utilise as a figure of merit for assessing the reconstructions. Figure 10.1 shows an example of this for 21 projections and 180 repetitions where the attenuation values for 10 pixels in the material ethanol have been plotted against the averaging of those 10 values. It is seen that the values are very volatile but are smoothed to form the energy curves.

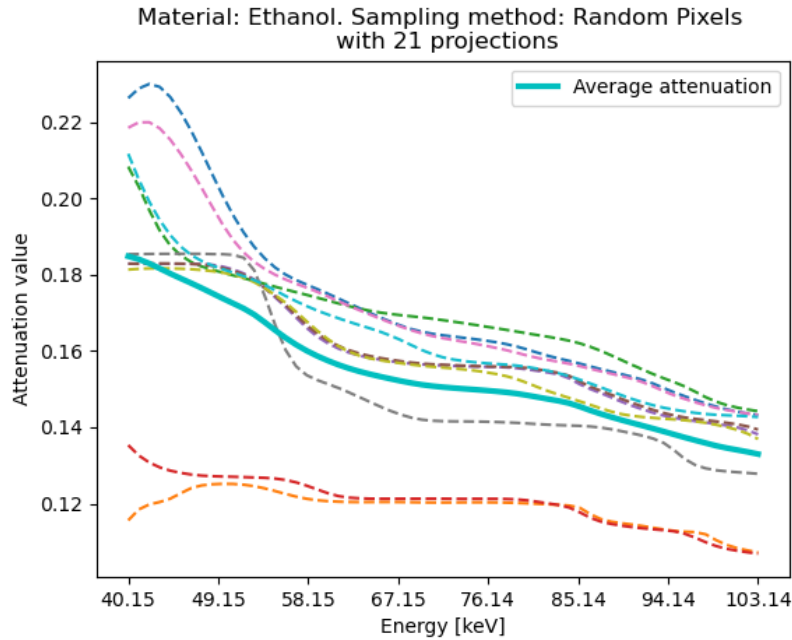


Figure 10.1: Attenuation values for 10 random sampled pixels in Ethanol for 21 projections and 120 repetitions. Dotted lines (- -) are different pixels in the material

Based on this could one argue that the method referred to as: *Error* $\rightarrow$ *Mean* is therefore a superior method for accurately capturing the true error between energy curves. This could maybe also be the reason that the plot 9.5b associated with *Random Pixels, error* has a more clearly seen minimum.

10.1.3 Best Acquisition Parameters

Both proposed figures of merit suggest the existence of a threshold for determining optimal acquisition parameters. The RMSE metric identifies 420 projections as optimal,

whereas the energy curves based error metric indicates that 36, 45, 70, and 90 projections yield optimal results. This is a large range but it can be said to be in consistent with the different focus areas of the two FOM. It is therefore reasonable to expect divergent outcomes between the two measures. Furthermore, it is crucial to emphasise that the obtained results are highly specific to the utilised data set. For instance, it is highly plausible that the optimal regularisation parameter α , are highly specific not only to the number of projections and repetitions but also to the overall scanning configuration.

10.1.4 Downsampling Technique

The downsampling of projections has been performed by selectively extracting every x -th projection based on the factorisation approach. For instance, to obtain 1260 projections, every second projection was sampled from the original data set. Although this method is straightforward to implement, it does not guarantee the uniform utilisation of identical angles across all downsampling instances. Consequently, if a particular angle is affected by external factors, leading to signal degradation, it is plausible that this angle may be included in certain downsampling instances but excluded in others. As a result, a certain degree of uncertainty arises when comparing results from one downsampling to another, primarily due to variations in the selected projections and their corresponding angles. Alternatively could golden-ratio approach for distributions of angles be used to downsample the data. The method was inspired by the way plants distribute their leaves [24]. The method increment projection angles using $\Delta\phi$, which is the golden ratio times 360°

$$\Delta\phi = \frac{\sqrt{5} - 1}{2} \cdot 360^\circ \approx 222.5^\circ \quad (10.1)$$

This method ensures that when additional angles sampled are the previously sampled angles are reused. Consequently, this approach minimises inconsistencies in the downsampling process, as all downsampling instances incorporate the same set of angles, including both the "bad" and "good" angles. By reusing previously sampled angles, the potential variations introduced by different sets of angles are mitigated, thereby promoting greater consistency and comparability across downsamplings.

10.2 Further work

10.2.1 Varying Photon Count

Though this project the photon count have been kept constant, thereby ensuring that the downsampling of the data set aligns with the prescribed combinations delineated in Table 8.3. The next natural step would be to extend the scope of the conducted experiments to include additional combinations wherein the photon count is not held constant. It could be interesting to explore whether the reconstruction quality exhibits a proportional improvement with an increase in the number of projections and acquisition

repetitions, or if a threshold is reached beyond which increasing either fails to yield a better reconstruction outcome. This investigation could further shed light on the interplay between projection quantity and repetition frequency to further the understanding of the optimal acquisition parameters governing the reconstruction process.

10.2.2 Regularisation in the Energy Dimension

An additional avenue for further exploration could be the continuation of the work started at the CIL hackaton, where a framework for using Tikhonov regularisation in the energy dimension was set up. Although this regularisation problem fell outside the initial scope of the project, it represents a promising direction for subsequent investigation. Given that we have proposed a figure of merit based on attenuation curves, is it particularly interesting to establish a connection between this figure of merit and the regularisation problem, which specifically focuses on the attenuation. Moreover, conducting an in-depth analysis to determine the optimal choice for the regularisation parameter β , would provide valuable insights into the regularisation process and its impact on the resulting energy curves.

This project set out to investigate the impact of different acquisition parameters influence on the efficacy of reconstruction quality. Specifically, the focus was placed on identifying and implementing distinct figures of merit (FOM) that could effectively assess the quality of reconstructions obtained from a 4D hyperhyperspectral data set. These figures of merit were designed to facilitate the identification of optimal acquisition parameters under the condition of a constant photon count. By employing such figures of merit, valuable insights could be collected, aiding in the determination of acquisition configurations that would yield optimal reconstruction outcomes.

In this study, two distinct figures of merit were proposed and employed to assess the quality of reconstructions. The first figure of merit used the Root Mean Square Error (RMSE) metric, which facilitated a direct comparison between the reconstructed images and a ground truth reference. This metric proved to be straightforward to implement and easy to interpret. The second figure of merit introduced a range of metrics founded on the analysis of attenuation curves within the reconstruction process. This figure of merit was particularly interesting to develop due to the nature of the data used being hyperspectral, and the classification of materials within the sample often being based on an analysis of attenuation curves. The metric was however complicated to implement due to various choices for both pixel sampling and error calculations. As a result, four closely related methods were devised, each based on the analysis of attenuation curves, to capture the nuances of the reconstructions.

A comprehensive investigation was undertaken to explore the determination of the optimal regularisation parameter, α , for the total variation regularisation term. The study was conducted by assessing the impact of different choices for α using the Root Mean Square Error (RMSE) as an error measure. The RMSE was selected for its simplicity and ease of implementation. This study was done using a small already existing data set. It was concluded that a full 'sweep' needed to be conducted for each downsampling scenario in order to find the optimal α . It was concluded that the existing data set did not consist of a sufficient amount of projections and acquisition repetitions, and it was therefore deemed essential for the further work to design and capture a new data set specifically fit to this project.

The two FOMs were used to quantify the performance of reconstructions under varying numbers of projections and repetitions. The investigation was limited to combinations where the photon count remained constant, and aimed to discern if a threshold existed beyond which increasing the number of projections failed to yield a significant improvement in reconstruction quality. The RMSE method identified 420 projections as the optimal value, which was visually confirmed to be a better reconstruction compared to higher projection counts. On the other hand, the energy curve-based methods determined that 36, 45, 70, and 90 projections yielded the highest quality reconstructions. Based on these findings, it can be concluded that the two proposed figures of merit

capture different characteristics of the reconstruction to determine the associated error. Both FOMs do still prove valuable measures, providing distinct insights into the reconstruction quality. Their effectiveness, however, may vary depending on the specific application and context. Therefore, it is reasonable to conclude that both figures of merit hold significance and can provide valuable information, each suited for different aspects and requirements within a given application.

The results of the acquisition experiments, employing both the root mean square error (RMSE) and energy curve based error metrics, does not yield a definitive conclusion regarding the optimal number of projections required for a reliable reconstruction. This discrepancy arises due to the significantly divergent outcomes obtained from the two distinct error measures. However, it is reasonable to conclude that an optimal combination of projections and repetitions does exist, and it can be determined by employing an appropriate figure of merit.

An additional objective of this project was also to contribute towards establishing a CIL workflow tailored specifically for hyperspectral X-ray projection analysis for future users to gather inspiration from. A Jupyter notebook summarising the work conducted during the hackathon was prepared with the intention of publicly sharing it. Furthermore have a complete data processing pipeline for hyperspectral data been developed spanning; loading of data, detector preprocessing, masking, ring artefacts removal, and regularisation set-up including parameter selection. The pipeline was developed using real data. All the Python code written throughout the course of this project be made accessible on GitHub. This will allow interested individuals to access and explore the code providing interested used inspiration on hyperspectral data analysis within the CIL framework https://github.com/IdaPug/Bachelor_project_2023.

This project was motivated by the issues surrounding airport baggage scanning, where it is essential to achieve swift scanning while simultaneously attaining satisfactory reconstructions. Specifically was it interesting to explore the optimal scan conditions for the number of projections and acquisition repetitions, ensuring high-quality reconstructions while maintaining a limited photon budget. Experiments were formulated that helped address this matter of the relationship between the number of projections and acquisition repetition, and the project successfully introduced figure of merits, facilitating the assessment of reconstruction quality in future baggage scanning experiments.

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